Edition 30.10.2020

$$\frac{\partial n_a}{\partial t}+\frac{1}{\sqrt{g}}\frac{\partial}{\partial x}\left(\frac{\sqrt{g}}{h_x}\tilde{\Gamma}_{ax}\right)+\frac{1}{\sqrt{g}}\frac{\partial}{\partial y}\left(\frac{\sqrt{g}}{h_y}\tilde{\Gamma}_{ay}\right)=S_a^n\text{, } a = 0,\ 1,\ldots,n_s\text{-}1$$

$$\tilde{\Gamma}_{ax}=\begin{cases}\left(b_xV_{\parallel a}+V_{ax}^{(E)}+V_{ax}^{(AN)}+V_a^{corr\_dpc}+0\cdot\dfrac{j_x^{(AN)}+j_x^{(in)}+\tilde{j}_x^{(vis\parallel)}+\tilde{j}_x^{(visq)}+j_x^{(vis\perp)}}{en_a}\right)n_a- \\ \qquad -D_a^n\dfrac{1}{h_x}\dfrac{\partial n_a}{\partial x}-\tilde{D}_{ax}^p\dfrac{1}{h_x}\dfrac{\partial p_a}{\partial x}-\tilde{\Gamma}_{ax}^{(PSch)}, \quad z_{n_a}=1\ \&\ z_a=1 \\ \left(b_xV_{\parallel a}+V_{ax}^{(E)}+V_{ax}^{(AN)}+V_a^{corr\_dpc}\right)n_a-D_a^n\dfrac{1}{h_x}\dfrac{\partial n_a}{\partial x}-\tilde{D}_{ax}^p\dfrac{1}{h_x}\dfrac{\partial p_a}{\partial x}-\tilde{\Gamma}_{ax}^{(PSch)}\end{cases}$$

$$\tilde{\Gamma}_{ay}=\begin{cases}\left(V_{ay}^{(E)}+V_{ay}^{(AN)}+\dfrac{j_y^{(AN)}+j_y^{(in)}+\tilde{j}_y^{(vis\parallel)}+\tilde{j}_y^{(visq)}}{en_a}\right)n_a- \\ \qquad -D_a^n\dfrac{1}{h_y}\dfrac{\partial n_a}{\partial y}-\tilde{D}_{ay}^p\dfrac{1}{h_y}\dfrac{\partial p_a}{\partial y}-\tilde{\Gamma}_{ay}^{(PSch)}+\dfrac{j_y^{(vis\perp)}}{e}, \quad z_{n_a}=1\ \&\ z_a=1\ , \\ \left(V_{ay}^{(E)}+V_{ay}^{(AN)}\right)n_a-D_a^n\dfrac{1}{h_y}\dfrac{\partial n_a}{\partial y}-\tilde{D}_{ay}^p\dfrac{1}{h_y}\dfrac{\partial p_a}{\partial y}-\tilde{\Gamma}_{ay}^{(PSch)},\end{cases}$$

$$\Gamma_{ax}^{(PSch)}=D_{axy}\frac{1}{h_y}\frac{\partial n_aT_i}{\partial y} \qquad \Gamma_{ay}^{(PSch)}=-D_{ayx}\frac{1}{h_x}\frac{\partial n_aT_i}{\partial x}$$

$$\tilde{\Gamma}_{ax}^{(PSch)}=\begin{cases}\Gamma_{ax}^{(PSch)}, & \left(z_{n_a}=1\wedge z_a\neq 1\right)\vee z_a=0 \\ \dfrac{\Gamma_{ax}^{(PSch)}}{1+\dfrac{\left|\Gamma_{ax}^{(PSch)}\right|}{\dfrac{\sqrt{g}}{h_x}n_aV_{T_a}}}, & \left(z_{n_a}\neq 1\vee z_a\neq 1\right)\wedge z_a\neq 0\end{cases} \qquad \tilde{\Gamma}_{ay}^{(PSch)}=\begin{cases}\Gamma_{ay}^{(PSch)}, & \left(z_{n_a}=1\wedge z_a\neq 1\right)\vee z_a=0 \\ \dfrac{\Gamma_{ay}^{(PSch)}}{1+\dfrac{\left|\Gamma_{ay}^{(PSch)}\right|}{\dfrac{\sqrt{g}}{h_y}n_aV_{T_a}}}, & \left(z_{n_a}\neq 1\vee z_a\neq 1\right)\wedge z_a\neq 0\end{cases}$$

$$V_{T_a}=|b_x|\sqrt{\frac{8T_i}{\pi m_a}}$$

$$\Gamma_{ax}=\begin{cases}\left(b_xV_{\parallel a}+\tilde{U}_{xa}^{(dia)}+V_{ax}^{(AN)}+V_a^{corr\_dpc}+0\cdot\dfrac{j_x^{(AN)}+j_x^{(in)}+\tilde{j}_x^{(vis\parallel)}+\tilde{j}_x^{(visq)}+j_x^{(vis\perp)}}{en_a}\right)n_a- \\ \qquad -D_a^n\dfrac{1}{h_x}\dfrac{\partial n_a}{\partial x}-\tilde{D}_{ax}^p\dfrac{1}{h_x}\dfrac{\partial p_a}{\partial x}, \quad z_{n_a}=1\ \&\ z_a=1 \\ \left(b_xV_{\parallel a}+\tilde{U}_{xa}^{(dia)}+V_{ax}^{(AN)}+V_a^{corr\_dpc}\right)n_a-D_a^n\dfrac{1}{h_x}\dfrac{\partial n_a}{\partial x}-\tilde{D}_{ax}^p\dfrac{1}{h_x}\dfrac{\partial p_a}{\partial x},\end{cases}$$

$$\Gamma_{ay} = \begin{cases} \left(\widetilde{U}_{ya}^{(dia)} + V_{ay}^{(AN)} + \dfrac{j_y^{(AN)} + j_y^{(in)} + \widetilde{j}_y^{(vis\|)} + \widetilde{j}_y^{(visq)}}{en_a} \right) n_a - D_a^n \dfrac{1}{h_y} \dfrac{\partial n_a}{\partial y} - \widetilde{D}_{ay}^p \dfrac{1}{h_y} \dfrac{\partial p_a}{\partial y} + \dfrac{j_y^{(vis\perp)}}{e}, & z_{n_a} = 1 \;\&\; z_a = 1 \\ \left(\widetilde{U}_{ya}^{(dia)} + V_{ay}^{(AN)} \right) n_a - D_a^n \dfrac{1}{h_y} \dfrac{\partial n_a}{\partial y} - \widetilde{D}_{ay}^p \dfrac{1}{h_y} \dfrac{\partial p_a}{\partial y} \end{cases}$$

$$\Gamma_{ax}^{Cor} = \begin{cases} \left(b_x V_{\|a} + \widetilde{U}_{xa}^{(2dia)} + V_{ax}^{(AN)} + V_a^{corr\_dpc} + 0 \cdot \dfrac{j_x^{(AN)} + j_x^{(in)} + \widetilde{j}_x^{(vis\|)} + \widetilde{j}_x^{(visq)} + j_x^{(vis\perp)}}{en_a} \right) n_a - \\ \qquad - D_a^n \dfrac{1}{h_x} \dfrac{\partial n_a}{\partial x} - \widetilde{D}_{ax}^p \dfrac{1}{h_x} \dfrac{\partial p_a}{\partial x}, & z_{n_a} = 1 \;\&\; z_a = 1 \\ \left(b_x V_{\|a} + \widetilde{U}_{xa}^{(2dia)} + V_{ax}^{(AN)} + V_a^{corr\_dpc} \right) n_a - D_a^n \dfrac{1}{h_x} \dfrac{\partial n_a}{\partial x} - \widetilde{D}_{ax}^p \dfrac{1}{h_x} \dfrac{\partial p_a}{\partial x} \end{cases}$$

$$\Gamma_{ay}^{Cor} = \begin{cases} \left(\widetilde{U}_{ya}^{(2dia)} + V_{ay}^{(AN)} + \dfrac{j_y^{(AN)} + j_y^{(in)} + \widetilde{j}_y^{(vis\|)} + \widetilde{j}_y^{(visq)}}{en_a} \right) n_a - D_a^n \dfrac{1}{h_y} \dfrac{\partial n_a}{\partial y} - \widetilde{D}_{ay}^p \dfrac{1}{h_y} \dfrac{\partial p_a}{\partial y} + \dfrac{j_y^{(vis\perp)}}{e}, & z_{n_a} = 1 \;\&\; z_a = 1 \\ \left(\widetilde{U}_{ya}^{(2dia)} + V_{ay}^{(AN)} \right) n_a - D_a^n \dfrac{1}{h_y} \dfrac{\partial n_a}{\partial y} - \widetilde{D}_{ay}^p \dfrac{1}{h_y} \dfrac{\partial p_a}{\partial y} \end{cases}$$

$$V_{ax}^{(E)} = \begin{cases} -\dfrac{B_z}{B^2} \dfrac{1}{h_y} \dfrac{\partial \phi}{\partial y}, & z_a \neq 0, \\ 0, & z_a = 0 \end{cases}, \qquad V_{ay}^{(E)} = \begin{cases} \dfrac{B_z}{B^2} \dfrac{1}{h_x} \dfrac{\partial \phi}{\partial x}, & z_a \neq 0, \\ 0, & z_a = 0 \end{cases},$$

$$\widetilde{V}_{xa}^{(dia)} = \begin{cases} \dfrac{T_i B_z}{Z_a e} \dfrac{\partial}{h_y \partial y} \left(\dfrac{1}{B^2} \right), & z_a \neq 0, \\ 0, & z_a = 0 \end{cases}, \qquad \widetilde{V}_{ya}^{(dia)} = \begin{cases} -\dfrac{T_i B_z}{Z_a e} \dfrac{\partial}{h_x \partial x} \left(\dfrac{1}{B^2} \right), & z_a \neq 0, \\ 0, & z_a = 0 \end{cases},$$

$$\widetilde{\widetilde{V}}_{xa}^{(dia)} = \begin{cases} \widetilde{V}_{xa}^{(dia)}, & \left(z_{n_a} = 1 \wedge z_a \neq 1 \right) \vee z_a = 0 \\ \dfrac{\widetilde{V}_{xa}^{(dia)}}{1 + \dfrac{\left| \widetilde{V}_{xa}^{(dia)} \right|}{V_{T_a}}}, & \left(z_{n_a} \neq 1 \vee z_a \neq 1 \right) \wedge z_a \neq 0 \end{cases} \qquad \widetilde{\widetilde{V}}_{ya}^{(dia)} = \begin{cases} \widetilde{V}_{ya}^{(dia)}, & \left(z_{n_a} = 1 \wedge z_a \neq 1 \right) \vee z_a = 0 \\ \dfrac{\widetilde{V}_{ya}^{(dia)}}{1 + \dfrac{\left| \widetilde{V}_{ya}^{(dia)} \right|}{V_{T_a}}}, & \left(z_{n_a} \neq 1 \vee z_a \neq 1 \right) \wedge z_a \neq 0 \end{cases}$$

$$\Gamma_{ax}^{(Eir)} = \begin{cases} \left(b_x V_{\|a} + V_{ax}^{(E)} + \widetilde{W}_{xa}^{(dia)} + V_{ax}^{(AN)} + V_a^{corr\_dpc} + 0 \cdot \dfrac{j_x^{(AN)} + j_x^{(in)} + \widetilde{j}_x^{(vis\|)} + \widetilde{j}_x^{(visq)} + j_x^{(vis\perp)}}{en_a} \right) n_a - \\ \qquad - D_a^n \dfrac{1}{h_x} \dfrac{\partial n_a}{\partial x}, & z_{n_a} = 1 \;\&\; z_a = 1 \\ \left(b_x V_{\|a} + V_{ax}^{(E)} + \widetilde{W}_{xa}^{(dia)} + V_{ax}^{(AN)} + V_a^{corr\_dpc} \right) n_a - D_a^n \dfrac{1}{h_x} \dfrac{\partial n_a}{\partial x} \end{cases}$$

$$\Gamma_{ay}^{(Eir)}=\begin{cases}\left(V_{ay}^{(E)}+\widetilde{W}_{ya}^{(dia)}+V_{ay}^{(AN)}+\dfrac{j_y^{(AN)}+j_y^{(in)}+\widetilde{j}_y^{(vis\|)}+\widetilde{j}_y^{(visq)}}{en_a}\right)n_a-D_a^n\dfrac{1}{h_y}\dfrac{\partial n_a}{\partial y}+\dfrac{j_y^{(vis\perp)}}{e}, & z_{n_a}=1\ \&\ z_a=1\\ \left(V_{ay}^{(E)}+\widetilde{W}_{ya}^{(dia)}+V_{ay}^{(AN)}\right)n_a-D_a^n\dfrac{1}{h_y}\dfrac{\partial n_a}{\partial y} & \end{cases}$$

where

$$W_{xa}^{(dia)}=\begin{cases}-\dfrac{B_z}{Z_aeB^2n_a}\dfrac{\partial n_aT_i}{h_y\partial y}, & z_a\neq 0,\\ 0, & z_a=0\end{cases}\qquad W_{ya}^{(dia)}=\begin{cases}\dfrac{B_z}{Z_aeB^2n_a}\dfrac{\partial n_aT_i}{h_x\partial x}, & z_a\neq 0,\\ 0, & z_a=0\end{cases}$$

$$\widetilde{W}_{xa}^{(dia)}=\begin{cases}W_{xa}^{(dia)}, & \left(z_{n_a}=1\wedge z_a\neq 1\right)\vee z_a=0\\ \dfrac{W_{xa}^{(dia)}}{1+\dfrac{\left|W_{xa}^{(dia)}\right|}{V_{T_a}}}, & \left(z_{n_a}\neq 1\vee z_a\neq 1\right)\wedge z_a\neq 0\end{cases}\qquad \widetilde{W}_{ya}^{(dia)}=\begin{cases}W_{ya}^{(dia)}, & \left(z_{n_a}=1\wedge z_a\neq 1\right)\vee z_a=0\\ \dfrac{W_{ya}^{(dia)}}{1+\dfrac{\left|W_{ya}^{(dia)}\right|}{V_{T_a}}}, & \left(z_{n_a}\neq 1\vee z_a\neq 1\right)\wedge z_a\neq 0\end{cases}$$

$$U_{xa}^{(dia)}=\widetilde{V}_{xa}^{(dia)}+V_{ax}^{(E)}\qquad U_{ya}^{(dia)}=\widetilde{V}_{ya}^{(dia)}+V_{ay}^{(E)}$$

$$U_{xa}^{(2dia)}=2\widetilde{V}_{xa}^{(dia)}+V_{ax}^{(E)}\qquad U_{ya}^{(2dia)}=2\widetilde{V}_{ya}^{(dia)}+V_{ay}^{(E)}$$

$$\widetilde{U}_{xa}^{(dia)}=\begin{cases}U_{xa}^{(dia)}, & \left(z_{n_a}=1\wedge z_a\neq 1\right)\vee z_a=0\\ \dfrac{U_{xa}^{(dia)}}{1+\dfrac{\left|U_{xa}^{(dia)}\right|}{V_{T_a}}}, & \left(z_{n_a}\neq 1\vee z_a\neq 1\right)\wedge z_a\neq 0\end{cases}\qquad \widetilde{U}_{ya}^{(dia)}=\begin{cases}U_{ya}^{(dia)}, & \left(z_{n_a}=1\wedge z_a\neq 1\right)\vee z_a=0\\ \dfrac{U_{ya}^{(dia)}}{1+\dfrac{\left|U_{ya}^{(dia)}\right|}{V_{T_a}}}, & \left(z_{n_a}\neq 1\vee z_a\neq 1\right)\wedge z_a\neq 0\end{cases}$$

$$\widetilde{U}_{xa}^{(2dia)}=\begin{cases}U_{xa}^{(2dia)}, & \left(z_{n_a}=1\wedge z_a\neq 1\right)\vee z_a=0\\ \dfrac{U_{xa}^{(2dia)}}{1+\dfrac{\left|U_{xa}^{(2dia)}\right|}{V_{T_a}}}, & \left(z_{n_a}\neq 1\vee z_a\neq 1\right)\wedge z_a\neq 0\end{cases}\qquad \widetilde{U}_{ya}^{(2dia)}=\begin{cases}U_{ya}^{(2dia)}, & \left(z_{n_a}=1\wedge z_a\neq 1\right)\vee z_a=0\\ \dfrac{U_{ya}^{(2dia)}}{1+\dfrac{\left|U_{ya}^{(2dia)}\right|}{V_{T_a}}}, & \left(z_{n_a}\neq 1\vee z_a\neq 1\right)\wedge z_a\neq 0\end{cases}$$

$$E_{k,a}=\frac{m_a}{2}u^2_{\|a}$$

Source terms in density equations

$$S_a^n=S_{I_a}^n+S_{I_{a+1}}^n+S_{R_a}^n+S_{R_{a-1}}^n+S_{CX\ a}^n+S_{CX\ a-1}^n,$$

z_a - the atomic charge of species (a)

z_{n_a} - the nuclear charge of species (a)

$$S^n_{I_a} = \begin{cases} -r_{I_a} n_e n_a, & a = 0,\ldots,n_s - 2, \quad z_a < z_{a+1} \quad and \quad z_{n_a} = z_{n_{a+1}} \\ 0, & \end{cases},$$

$$S^n_{I_{a+1}} = \begin{cases} r_{I_a} n_e n_a, & a = 0,\ldots,n_s - 2, \; z_a < z_{a+1} \quad and \quad z_{n_a} = z_{n_{a+1}} \\ 0 & \end{cases}$$

$$r_{I_a} = \exp\left(r_{0,I_a} + r_{1,I_a} \ln\left(\frac{T_e}{e}\right)\right)$$

$$S^n_{R_a} = \begin{cases} -r_{R_a} n_e n_a, & a = 1,\ldots,n_s - 1, \; z_{a-1} < z_a \quad and \quad z_{n_{a-1}} = z_{n_a} \\ 0 & \end{cases}$$

$$S^n_{R_{a-1}} = \begin{cases} r_{R_a} n_e n_a, & a = 1,\ldots,n_s - 1, \; z_{a-1} < z_a \quad and \quad z_{n_{a-1}} = z_{n_a} \\ 0, & \end{cases},$$

$$r_{R_a} = \exp\left(r_{0,R_a} + r_{1,R_a} \ln\left(\frac{T_e}{e}\right)\right)$$

$$S^n_{CX_a} = \begin{cases} -r_{CX_a} n_{a_0} n_a, & a = 1,\ldots,n_s - 1, \; z_{a-1} < z_a \quad and \quad z_{n_{a-1}} = z_{n_a} \\ 0 & \end{cases} \qquad ????$$

$$S^n_{CX_{a-1}} = \begin{cases} r_{CX_a} n_{a_0} n_a, & a = 1,\ldots,n_s - 1, \; z_{a-1} < z_a \quad and \quad z_{n_{a-1}} = z_{n_a} \\ 0, & \end{cases} \qquad ?????$$

$$r_{CX_a} = \exp\left(r_{0,CX_a} + r_{1,CX_a} \ln\left(\frac{m_p T_i}{m_{a_0} e}\right)\right)$$

$$n_e = \sum_{a=0}^{n_s-1} z_a n_a, \qquad n_i = \sum_{a=0}^{n_s-1} n_a$$

$$p_a = n_a\left(z_a T_e + T_i\right), \qquad p_e = n_e T_e$$

$$V_a^{corr\_dpc} = \frac{c^{corr\_dpc}}{16} |b_x| \sqrt{\frac{p_a}{m_a n_a} \frac{\partial^2}{\partial x^2}\left(\frac{1}{p_a}\frac{\partial p_a}{\partial x}\right)}$$

$$\begin{cases} D_{axy} = \left(\frac{1}{B^2} - \frac{1}{\langle B^2 \rangle}\right)\frac{B_z}{z_a e}, & z_a \neq 0 \\ D_{axy} = 0, & z_a = 0 \end{cases}, \text{ where } \langle B^2 \rangle = \frac{\iint_{core} B^2 \sqrt{g}\, dxdy}{\iint_{core} \sqrt{g}\, dxdy}$$

$$\begin{cases} D_{ayx} = \left(\frac{1}{B^2} - \frac{1}{\langle B^2 \rangle}\right)\frac{B_z}{z_a e}, & z_a \neq 0 \\ D_{ayx} = 0, & z_a = 0 \end{cases}$$

$$D^n_a = D^n_{ANa}$$

$$D_{ANa}^{n} = \left(c_{D_a^N,0} + c_{D_a^N,1}\frac{10^{20}}{n_e} + c_{D_a^N,2} D_a^f + c_{D_a^N,3}\frac{6{,}2\cdot 10^{-2} T_e}{e|B|} \right) f_D(x,y),$$

$$f_D(x,y) = \begin{cases} 1, & z_a = 0 \\ \alpha_{1,D}, & -\delta_{1,D} \le y \le 0,\ x \in Core,\ z_a \ne 0 \\ \alpha_{2,D}, & 0 \le y \le \delta_{2,D},\ x \in Core, \\ \alpha_{3,D}, & x, y \in \Pr \cup SOL \end{cases}$$

$$a = 0,\ c_{D_a^N,0} = 0,\ c_{D_a^N,1} = 0,\ c_{D_a^N,2} = 0,\ c_{D_a^N,3} = 0$$

$$a = 1,\ c_{D_a^N,0} = 0,\ c_{D_a^N,1} = 0,\ c_{D_a^N,2} = 1,\ c_{D_a^N,3} = 0$$

$$D_a^f = c_{D_a^f,0} + c_{D_a^f,1}\frac{10^{20}}{n_i} + \frac{c_{D_a^f,2}\sqrt{\dfrac{T_i}{m_a}}}{c_{D_a^f,3} n_i + c_{D_a^f,4} n_e},$$

a	$c_{D_a^f,0}$	$c_{D_a^f,1}$	$c_{D_a^f,2}$	$c_{D_a^f,3}$	$c_{D_a^f,4}$	$c_{D_a^f,5}$	$c_{D_a^f,6}$	$c_{D_a^f,7}$
0	10^2	0	1	$5\cdot 10^{-20}$	$1{,}5\cdot 10^{-18}$	0	0	0
1	0,3	0	0	–	–	0	0	0

$$\tilde{D}_{ax}^{p} = \begin{cases} \dfrac{D_a^p}{\left(1 + \left(\dfrac{\dfrac{\sqrt{g}}{h_x} D_a^p \left|\dfrac{1}{h_x}\dfrac{\partial p_a}{\partial x}\right|}{\alpha \dfrac{\sqrt{g}}{h_x} n_a \dfrac{V_{T_i}}{4}}\right)^{\gamma_1}\right)^{\frac{1}{\gamma_1}}}, & V_{T_i} = \sqrt{\dfrac{8T_i}{\pi m_a}},\ z_a = 0 \\ D_a^p, & z_a \ne 0 \end{cases}$$

$$\tilde{D}_{ay}^{p} = \begin{cases} \dfrac{D_a^p}{\left(1 + \left(\dfrac{\dfrac{\sqrt{g}}{h_y} D_a^p \left|\dfrac{1}{h_y}\dfrac{\partial p_a}{\partial y}\right|}{\alpha \dfrac{\sqrt{g}}{h_y} n_a \dfrac{V_{T_i}}{4}}\right)^{\gamma_1}\right)^{\frac{1}{\gamma_1}}}, & V_{T_i} = \sqrt{\dfrac{8T_i}{\pi m_a}},\ z_a = 0 \\ D_a^p, & z_a \ne 0 \end{cases}$$

$$D_a^p = D_{ANa}^p$$

$$D_{ANa}^{p} = \frac{1}{z_a T_e + T_i}\left(c_{D_a^p,0} + c_{D_a^p,1}\frac{10^{20}}{n_e} + c_{D_a^p,2} D_a^f + c_{D_a^p,3}\frac{6{,}2\cdot 10^{-2} T_e}{e|B|} \right),$$

$$a = 0,\ c_{D_a^p,0} = 0,\ c_{D_a^p,1} = 0,\ c_{D_a^p,2} = 1,\ c_{D_a^p,3} = 0$$

$$a = 1,\ c_{D_a^p,0} = 0,\ c_{D_a^p,1} = 0,\ c_{D_a^p,2} = 0,\ c_{D_a^p,3} = 0$$

$$V_{ax}^{(AN)} = 0$$

$$V_{ay}^{(AN)} = c_{V_a^{(AN)},0} + c_{V_a^{(AN)},1}\frac{10^{20}}{n_e} + c_{V_a^{(AN)},2}D_a^f + c_{V_a^{(AN)},3}\frac{6{,}2\cdot 10^{-2}\,T_e}{e|B|}$$

$$a = 0,\ c_{V_a^{(AN)},0} = 0,\ c_{V_a^{(AN)},1} = 0,\ c_{V_a^{(AN)},2} = 0,\ c_{V_a^{(AN)},3} = 0$$

$$a = 1,\ c_{V_a^{(AN)},0} = 0,\ c_{V_a^{(AN)},1} = 0,\ c_{V_a^{(AN)},2} = 0,\ c_{V_a^{(AN)},3} = 0$$

New form

$$m_a \frac{\partial n_a V_{\|a}}{\partial t} + \frac{1}{h_z\sqrt{g}}\frac{\partial}{\partial x}\left(\frac{h_z\sqrt{g}}{h_x}\Gamma_{ax}^m\right) + \frac{1}{h_z\sqrt{g}}\frac{\partial}{\partial y}\left(\frac{h_z\sqrt{g}}{h_y}\Gamma_{ay}^m\right) + \frac{b_x}{h_x}\frac{\partial n_a T_i}{\partial x} + Z_a e n_a \frac{b_x}{h_x}\frac{\partial \varphi}{\partial x} =$$

$$= S_{a\|}^m + S_{CF_a}^m + S_{fr_a}^m + S_{Term\,a}^m + S_{I_a}^m + S_{R_a}^m + S_{CX_a}^m + S_{AN_a}^m$$

Old form

$$m_a \frac{\partial n_a V_{\|a}}{\partial t} + \frac{1}{h_z\sqrt{g}}\frac{\partial}{\partial x}\left(\frac{h_z\sqrt{g}}{h_x}\Gamma_{ax}^m\right) + \frac{1}{h_z\sqrt{g}}\frac{\partial}{\partial y}\left(\frac{h_z\sqrt{g}}{h_y}\Gamma_{ay}^m\right) + \frac{b_x}{h_x}\frac{\partial n_a T_i}{\partial x} + \frac{b_x}{h_x}\frac{Z_a n_a \partial n_e T_e}{n_e \partial x} =$$

$$= S_{a\|}^m + S_{CF,a}^m + S_{fr_a}^m + S_{Term\,a}^m + S_{I_a}^m + S_{R_a}^m + S_{CX_a}^m \qquad \textit{old form}$$

$$S_{a\|}^m = -\left(\nabla \pi_a^{\|}\right)_{\|}$$

$$\left(\nabla \pi_a^{\|}\right)_{\|} = \begin{cases} 0, \\ -\dfrac{\partial}{h_z\sqrt{g}\,\partial x}\left(\dfrac{h_z\sqrt{g}}{h_x}\dfrac{4}{3}\tilde{\tilde{\eta}}_{CLax}\dfrac{\partial \ln\left(h_z B^{\frac{1}{2}}\right)}{B^{\frac{1}{2}} h_x \partial x}\right) B^{\frac{1}{2}} V_{\|a} - B^{\frac{3}{2}} b_x \dfrac{\partial}{h_x \partial x}\left(\alpha^{NEO}\dfrac{b_x \tilde{\tau}_a}{B^2}\dfrac{\partial\left(B^{\frac{1}{2}}\tilde{q}_a^{(0)}\right)}{h_x \partial x}\right), & z_{n_a} = 1 \;\&\; z_a = 1 \end{cases}$$

$$S_{CF,a}^m = m_a b_x n_a V_{\|a}^2 \frac{1}{h_z h_x}\frac{\partial h_z}{\partial x}$$

$$S_{AN_a}^m = \begin{cases} 0, & z_a \neq 1 \;\; or \;\; z_{n,a} \neq 1 \\ -j_y^{(AN)}\dfrac{BB_x}{B_z}, & z_a = 1 \;\; and \;\; z_{n,a} = 1 \end{cases}$$ -correcting source due to artificial anomalous electric conductivity

$$S_a^m = S_{fr,a}^m + S_{Term}^m + S_{I_a}^m + S_{R_a}^m + S_{CX_a}^m,$$

Parallel momentum source due to ionization

$$S_{I_a}^m = S_{I_a}^q + S_{I_{a+1}}^q$$

$$S_{I_a}^q = \begin{cases} -r_{I_a} n_e n_a \left(\dfrac{m_a + m_{a+1}}{2}\right) V_{\|a}, & a = 0,\dots,n_s - 2, \quad z_a < z_{a+1} \;\; and \;\; z_{n_a} = z_{n_{a+1}} \\ 0, \end{cases},$$

$$S_{I_{a+1}}^q = \begin{cases} r_{I_a} n_e n_a \left(\dfrac{m_a + m_{a+1}}{2}\right) V_{\|a}, & a = 0,\dots,n_s - 2, \quad z_a < z_{a+1} \;\; and \;\; z_{n_a} = z_{n_{a+1}} \\ 0, \end{cases},$$

Parallel momentum source due to recombination

$$S_{R_a}^m = S_{R_a}^q + S_{R_{a-1}}^q$$

$$S^q_{R_a} = \begin{cases} -r_{R_a} n_e n_a \left(\dfrac{m_a + m_{a-1}}{2} \right) V_{\| a}, & a = 1,...,n_s - 1,\ z_{a-1} < z_a \quad and \quad z_{n_{a-1}} = z_{n_a} \\ 0 \end{cases}$$

$$S^q_{R_{a-1}} = \begin{cases} r_{R_a} n_e n_a \left(\dfrac{m_a + m_{a-1}}{2} \right) V_{\| a}, & a = 1,...,n_s - 1,\ z_{a-1} < z_a \quad and \quad z_{n_{a-1}} = z_{n_a} \\ 0, \end{cases},$$

$$S^m_{CX_a} =$$

New friction

$$S^m_{fr_a} = S^m_{fr,ea} + S^m_{fr,ia}$$

$$S^m_{fr,ea} = \frac{b_x^2 e^2 z_a^2}{\sigma_{\|} z_{eff}} n_e n_a \left(V_{e\|} - V_{a\|} \right) \qquad z_a \neq 0$$

$$S^m_{fr,ia} = \sum_{\substack{b=0 \\ b \neq a}}^{n_s-1} R_{ab}^{(V_\|)} = \sum_{\substack{b=0 \\ b \neq a}}^{n_s-1} \frac{1}{\zeta_p} m_p \sqrt{\frac{\dfrac{m_a}{m_p} \dfrac{m_b}{m_p}}{\dfrac{m_a}{m_p} + \dfrac{m_b}{m_p}}} z_a^2 z_b^2 n_a n_b K_{ab}^{(V_\|)} \left(V_{b\|} - V_{a\|} \right)$$

$$\zeta_p = \frac{3}{4\sqrt{2\pi}} \frac{\sqrt{m_p} T_i^{\frac{3}{2}}}{\Lambda} \left(\frac{4\pi\varepsilon_0}{e^2} \right)^2 \qquad\qquad \tau_a = \frac{\zeta_p}{z_a^2 z_{eff} n_e} \sqrt{\frac{m_a}{m_p}}$$

$$K_{rs}^{(V_\|)} = \begin{cases} c_{imp}^{(1)}, & r = r_{main},\ s \neq r, \quad z_r \neq 0,\ z_s \neq 0, \\ c_{imp}^{(1)}, & s = s_{main},\ r \neq s, \quad z_r \neq 0,\ z_s \neq 0, \\ 1, & r \neq r_{main},\ s \neq s_{main},\ z_r \neq 0,\ z_s \neq 0, \\ 0, & r = s, \quad or \quad z_r = 0,\ z_s = 0, \end{cases}$$

$$c_{imp}^{(1)} = \frac{(1 + 0.24 z_{eff\_imp})(1 + 0.93 z_{eff\_imp})}{(1 + 2.56 z_{eff\_imp})(1 + 0.29 z_{eff\_imp})} \qquad\qquad c_{imp}^{(2)} = 1.56 \frac{(1 + 1.4 z_{eff\_imp})(1 + 0.52 z_{eff\_imp})}{(1 + 2.56 z_{eff\_imp})(1 + 0.29 z_{eff\_imp})}$$

$$z_{eff\_imp} = \frac{\sum_{\substack{a=0 \\ a \neq a_{main}}}^{n_s-1} n_a z_a^2}{n_{a_{main}} z_{a_{main}}^2}$$

Old friction

$$S^m_{fr,a} = \sum_{\substack{b=0 \\ b \neq a, z_a \neq 0, z_b \neq 0}}^{n_s-1} \frac{1}{\tau_p} m_p \sqrt{\frac{\dfrac{m_a}{m_p} \dfrac{m_b}{m_p}}{\dfrac{m_a}{m_p} + \dfrac{m_b}{m_p}}} z_a^2 z_b^2 n_b n_a \left(V_{b\|} - V_{a\|} \right),$$ - friction force (b2sifr)

$$\tau_p = \frac{3}{4\sqrt{2\pi}} \frac{\sqrt{m_p} T_i^{\frac{3}{2}}}{\Lambda} \left(\frac{4\pi\varepsilon_0}{e^2} \right)^2$$, where the Coulomb logarithm Λ can be chosen as

$$\Lambda = \begin{cases} 12 \\ \begin{cases} \max\left\{5;\ 23{,}4 - 1{,}15\cdot\log_{10}\left(\dfrac{n_e}{10^6}\right) + 3{,}45\cdot\log_{10}\left(\dfrac{T_e}{e}\right)\right\}, & \dfrac{T_e}{e} \le 50 \\ \max\left\{5;\ 25{,}3 - 1{,}15\cdot\log_{10}\left(\dfrac{n_e}{10^6}\right) + 2{,}30\cdot\log_{10}\left(\dfrac{T_e}{e}\right)\right\}, & \dfrac{T_e}{e} > 50 \end{cases} \quad \text{Braginsky formula} \\ \max\left\{5;\ 17{,}3 - 0{,}5\cdot\log_e\left(\dfrac{n_e}{10^{20}}\right) + 1{,}5\cdot\log_e\left(\dfrac{T_e}{e\cdot 10^3}\right)\right\} \quad \text{Wesson formula for ion - ion collisions} \end{cases},$$

New thermal force

$$S^m_{Term\,a} = S^m_{Term,ea} + S^m_{Term,ia}$$

$$S^m_{Term,ea} = -\alpha_{ex} c_e^{(1)} \frac{n_a z_a^2 m_e}{b_x e \zeta_e} \frac{1}{c^{(Luciani)}_{\sigma_{CL\parallel}}} \frac{1}{h_x} \frac{\partial T_e}{\partial x}, \quad z_a \ne 0$$

$$S^m_{Term,ia} = \sum_{\substack{b=0 \\ b\ne a,}}^{n_s-1} c^{(F\lim)}_{\alpha_{ab}} z_a^2 z_b^2 \frac{m_p}{\zeta_p e} \frac{n_a n_b}{n_a + n_b} \alpha_{ab} \frac{1}{h_x} \frac{\partial T_i}{\partial x}, \quad z_a \ne 0$$

$$\alpha_{ab} = \frac{\zeta_p e (n_a + n_b)}{\sqrt{m_p}} \sqrt{\frac{\dfrac{m_a}{m_p}\dfrac{m_b}{m_p}}{\dfrac{m_a}{m_p} + \dfrac{m_b}{m_p}}} \left(\frac{K^{(Term)}_{ab}}{\kappa_b} - \frac{K^{(Term)}_{ba}}{\kappa_a}\right) b_x, \quad z_a \ne 0$$

$$\zeta_e = \frac{3}{4\sqrt{2\pi}} \frac{\sqrt{m_e} T_e^{\frac{3}{2}}}{\Lambda} \left(\frac{4\pi\varepsilon_0}{e^2}\right)^2 \qquad \tau_e = \frac{\zeta_e}{z_{eff} n_e}$$

$$c_e^{(1)} = \frac{(1 + 0.24 z_{eff})(1 + 0.93 z_{eff})}{(1 + 2.56 z_{eff})(1 + 0.29 z_{eff})}, \qquad c_e^{(2)} = 1.56 \frac{(1 + 1.4 z_{eff})(1 + 0.52 z_{eff})}{(1 + 2.56 z_{eff})(1 + 0.29 z_{eff})} \qquad \tilde{c}_e^{(2)} = \frac{c_e^{(2)}}{\left(z_{eff} + \dfrac{\sqrt{2}}{2}\right)}$$

$$K^{(Term)}_{rs} = \begin{cases} c^{(2)}_{imp}, & s \ne r, \quad s = s_{main}, \quad z_r \ne 0,\ z_s \ne 0, \\ 0, & r \ne r_{main}, \ s \ne s_{main}, \ z_r \ne 0,\ z_s \ne 0, \\ 0, & r = r_{main}, \ s \text{ - arbitrary} \end{cases}$$

$$\kappa_a = \sum_{r=0}^{n_s-1} z_a^2 z_r^2 n_r \sqrt{m_p} \sqrt{\frac{\dfrac{m_a}{m_p}\dfrac{m_r}{m_p}}{\dfrac{m_a}{m_p} + \dfrac{m_r}{m_p}}}$$

$$c^{(F\lim)}_{\alpha_{ab}} = \begin{cases} 1, & \text{inside the separatrix} \\ \dfrac{1}{1 + \dfrac{\left|\alpha_{ab} \dfrac{1}{h_x} \dfrac{\partial T_i}{\partial x}\right|}{c_{fl\lim ab}\, j_{ab\max}}}, & \text{outside the separatrix} \end{cases} \qquad j_{ab\max} = e K^{(V_\parallel)}_{ab} \sqrt{\frac{T_i}{m_p}} (n_a + n_b)$$

New coefficient $c_{fl\lim ab}$ is stored in cflim(4)

Old thermal force

$$S^m_{Term,a} = n_a\left(\frac{z_a}{n_e} - \frac{z_a^2}{n_{e^2}}\right)n_e b_x e\left[\hat{E}_x - \frac{1}{e}\left(c_{th_e,a}\Delta_{T_e} + c_{th_i,a}\Delta_{T_i}\right)\right],$$ thermal force (b2sifr)

$$n_{e^2} = \sum_{a=0}^{n_s-1} z_a^2 n_a, \qquad l_{t_e} = 0{,}3;\ \ l_{t_i} = 0{,}3;\ \ z_{\lambda_e} = \frac{1.5\cdot 10^{16}}{n_e}\left(\frac{T_e}{e}\right)^2, \qquad z_{\lambda_i} = \frac{1.5\cdot 10^{16}}{n_{e^2}}\left(\frac{T_i}{e}\right)^2$$

$$\Delta_{T_e} = \begin{cases} \dfrac{\partial T_e}{h_x\partial x}, & l_{t_e} \le 0 \\ \left|\min\left\{\left|\dfrac{\partial T_e}{h_x\partial x}\right|, \dfrac{l_{t_e}T_e}{z_{\lambda_e}|b_x|}\right\}\right|, & \dfrac{\partial T_e}{h_x\partial x} \ge 0,\quad l_{t_e} > 0 \\ -\left|\min\left\{\left|\dfrac{\partial T_e}{h_x\partial x}\right|, \dfrac{l_{t_e}T_e}{z_{\lambda_e}|b_x|}\right\}\right|, & \dfrac{\partial T_e}{h_x\partial x} < 0,\quad l_{t_e} > 0 \end{cases}$$

$$\Delta_{T_i} = \begin{cases} \dfrac{\partial T_i}{h_x\partial x}, & l_{t_i} \le 0 \\ \left|\min\left\{\left|\dfrac{\partial T_i}{h_x\partial x}\right|, \dfrac{l_{t_i}T_i}{z_{\lambda_i}|b_x|}\right\}\right|, & \dfrac{\partial T_i}{h_x\partial x} \ge 0,\quad l_{t_i} > 0 \\ -\left|\min\left\{\left|\dfrac{\partial T_i}{h_x\partial x}\right|, \dfrac{l_{t_i}T_i}{z_{\lambda_i}|b_x|}\right\}\right|, & \dfrac{\partial T_i}{h_x\partial x} < 0,\quad l_{t_i} > 0 \end{cases}$$

$c_{th_e,a} = 0,\ a = 0, n_s - 1$ and $c_{th_i,a} = 2{,}65,\ \ a = 0, n_s - 1$ if 21 moment approximation is not used.

$$\Gamma^m_{ax} = \begin{cases} m_a V_{\|a}\Gamma^{Cor}_{ax} + \dfrac{4}{3}\tilde{\tilde{\eta}}_{CLax}\dfrac{\partial \ln h_z}{h_x\partial x}V_{\|a} - \eta_{ax}\dfrac{\partial V_{\|a}}{h_x\partial x}, & z_{n_a} = 1\ \&\ z_a = 1 \\ m_a V_{\|a}\Gamma^{Cor}_{ax} - \eta_{ax}\dfrac{\partial V_{\|a}}{h_x\partial x}, & \end{cases}$$

$$\Gamma^m_{ay} = m_a V_{\|a}\Gamma^{Cor}_{ay} - \eta_{ay}\frac{\partial V_{\|a}}{h_y\partial y},$$

$$\eta_{ax} = \eta_{ANa} + \frac{4}{3}\tilde{\tilde{\eta}}_{CLax}$$

$$\eta_{ANa} = \begin{cases} \left(c_{\eta_a,0} + c_{\eta_a,1}\dfrac{10^{20}}{n_e} + c_{\eta_a,2}D^f_a + c_{\eta_a,3}\dfrac{6{,}2\cdot 10^{-2}T_e}{e|B|}\right)m_a n_a f_{\eta_a}(x,y), & z_a \ne 0 \\ \left(c_{\eta_a,0} + c_{\eta_a,1}\dfrac{10^{20}}{n_e} + c_{\eta_a,2}D^f_a + c_{\eta_a,3}\dfrac{6{,}2\cdot 10^{-2}T_e}{e|B|}\right)m_a n_a, & z_a = 0 \end{cases}$$

$$f_{\eta_a}(x,y)=\begin{cases}1, & z_a=0\\ \alpha_{1,\eta_a}, & -\delta_{1,\eta_a}\le y\le 0,\ x\in Core,\ z_a\ne 0\\ \alpha_{2,\eta_a}, & 0\le y\le \delta_{2,\eta_a},\ x\in Core,\\ \alpha_{3,\eta_a}, & x,y\in \Pr\cup SOL\end{cases}$$

$$a=0,\ c_{\eta_a,0}=0,\ c_{\eta_a,1}=0,\ c_{\eta_a,2}=1,\ c_{\eta_a,3}=0$$

$$a=1,\ c_{\eta_a,0}=0,\ c_{\eta_a,1}=0,\ c_{\eta_a,2}=1,\ c_{\eta_a,3}=0$$

$$\tilde{\tilde{\eta}}_{CLax}=c_{\eta_{ax}^{(CL)}}\tilde{\eta}_{CLax}$$

$$c_{\eta_{ax}^{(CL)}}=\begin{cases}1, & z_a\ne 0 \textit{ and only for inner flux surfaces}\\ \dfrac{1}{1+\dfrac{\frac{\sqrt{g}}{h_x}\tilde{\eta}_{CLax}\left|\frac{\partial V_{\|a}}{h_x\partial x}\right|}{V_{m_a}^{(\max)}}}, & \textit{for all others cases}\end{cases}$$

$$V_{m_a}^{(\max)}=c_{f\lim_2}\frac{\sqrt{g}}{h_x}|b_x|n_aT_i$$

$$\tilde{\eta}_{CLax}=c_{\eta_{ax}^{(CL)}}^{(Luciani)}\eta_{CLax}$$

‘`b2trcl_lluciani`’ ‘1’

$$c_{\eta_{ax}^{(CL)}}^{(Luciani)}=\begin{cases}\dfrac{1}{1+\dfrac{7{,}5\cdot 10^{16}\left(\frac{T_i}{e}\right)^2}{c(y)\sum\limits_{a=0}^{n_s-1}z_a^2n_a}}, & z_a\ne 0 \textit{ and only for inner flux surfaces,}\\ 1, & \textit{for all others cases}\end{cases}$$

For plateau and banana correction ‘`b2trcl_lluciani`’ ‘3’

$$c_{\eta_{ax}^{(CL)}}^{(Luciani)}=\begin{cases}\dfrac{1}{1+\dfrac{15{,}12\cdot 10^{16}\left(\frac{T_i}{e}\right)^2}{c(y)\sum\limits_{a=0}^{n_s-1}z_a^2n_a}}\cdot\dfrac{1}{1+\dfrac{15{,}12\cdot 10^{16}\cdot\varepsilon^{3/2}\left(\frac{T_i}{e}\right)^2}{c(y)\sum\limits_{a=0}^{n_s-1}z_a^2n_a}}, & z_a\ne 0 \textit{ and only for inner flux surfaces,}\\ 1, & \textit{for all others cases}\end{cases}$$

$$c(y)=\oint h_x|b_x|^{-1}dx, \qquad \varepsilon=\frac{\left(\oint h_x dx\right)^2}{\oint\sqrt{g}/h_y dx}$$

$$\eta_{CLax} = \begin{cases} 0, & z_a = 0 \\ \eta\sqrt{2}T_i n_a \tau_a b_x^2, & z_a \neq 0 \end{cases}, \ \eta = 0.96,$$

Old expression was $\eta_{CLax} = \begin{cases} 0, & z_a = 0 \\ \eta T_i n_a \tau_a b_x^2, & z_a \neq 0 \end{cases}, \ \eta = 1{,}357,$

$$\tau_a = \frac{3}{4\sqrt{2\pi}} \frac{\sqrt{m_a} T_i^{\frac{3}{2}}}{z_a^2 z_{eff} n_e \Lambda} \left(\frac{4\pi\varepsilon_0}{e^2} \right)^2$$, where the Coulomb logarithm Λ can be chosen as

$$\Lambda = \begin{cases} 12 \\ \begin{cases} \max\left\{5;\ 23{,}4 - 1{,}15 \cdot \log_{10}\left(\frac{n_e}{10^6}\right) + 3{,}45 \cdot \log_{10}\left(\frac{T_e}{e}\right)\right\}, & \frac{T_e}{e} \le 50 \\ \max\left\{5;\ 25{,}3 - 1{,}15 \cdot \log_{10}\left(\frac{n_e}{10^6}\right) + 2{,}30 \cdot \log_{10}\left(\frac{T_e}{e}\right)\right\}, & \frac{T_e}{e} > 50 \end{cases} & \text{Braginsky formula} \\ \max\left\{5;\ 17{,}3 - 0{,}5 \cdot \log_e\left(\frac{n_e}{10^{20}}\right) + 1{,}5 \cdot \log_e\left(\frac{T_e}{e \cdot 10^3}\right)\right\} & \text{Wesson formula for ion-ion collisions} \end{cases},$$

$$\eta_{ay} = \eta_{ANa} + \eta_{CLay}, \qquad \eta_{ay}^{(CL)} = 0$$

$$\tilde{q}_a^{(0)} = q^{(1)} arctg\left(\frac{q_a^{(0)}}{q^{(1)}} \right), \ \ q^{(1)} = c_{q1} n_i T_i \sqrt{\frac{T_i + T_e}{m_i}}$$

$c_{q1} = 0.5$ or `{'b2siav_cqip1' '0.1'}` in b2mn.dat

the three-point smoothing procedure is imposed twice on $\tilde{q}^{(0)}$ along x-direction

in case if `{'b2siav_style_qip' '0'}` in b2mn.dat

$$q_a^{(0)} = q_{a\|}^{(0)} + \frac{B}{B_x} q_{ax}^{(dia)} = \begin{cases} 0 & \textit{outside the separatrix} \\ -\frac{5}{2} \frac{B_z B}{B_x < B^2 >} \frac{n_a T_i}{e} \frac{\partial T_i}{h_y \partial y} & \textit{inside the separatrix} \end{cases}$$

in case if `{'b2siav_style_qip' '1'}` in b2mn.dat

$$q_a^{(0)} = q_{a\|}^{(0)} + \frac{B}{B_x} q_{ax}^{(dia)} = \begin{cases} 0 & \textit{outside the separatrix} \\ -\tilde{\tilde{k}}_{ix}^{(CL)} b_x \frac{\partial T_i}{h_x \partial x} - \frac{5}{2} \frac{B_z B}{B_x B^2} \frac{n_a T_i}{e} \frac{\partial T_i}{h_y \partial y} & \textit{inside the separatrix} \end{cases}$$

$$\tilde{\tau}_a = \frac{\tau_a^{(Br)}}{1 + \frac{15{,}12 \cdot 10^{16} \left(\frac{T_i}{e} \right)^2}{c(y) \sum_{a=0}^{n_s - 1} z_a^2 n_a}} \cdot \frac{1}{1 + \frac{15{,}12 \cdot 10^{16} \cdot \varepsilon^{3/2} \left(\frac{T_i}{e} \right)^2}{c(y) \sum_{a=0}^{n_s - 1} z_a^2 n_a}} \qquad \tau_a^{(Br)} = \frac{3}{4\sqrt{\pi}} \frac{\sqrt{m_a} T_i^{\frac{3}{2}}}{z_a^2 z_{eff} n_e \Lambda} \left(\frac{4\pi\varepsilon_0}{e^2} \right)^2$$

This correction is compatible **only** with plateau and banana correction ‘`b2trcl_lluciani`’ ‘3’

$\alpha^{NEO} = \frac{8}{15}\left(\frac{-0.17+1.05\nu_*^{1/2}+2.7\nu_*^2\varepsilon^3}{1+0.7\nu_*^{1/2}+\nu_*^2\varepsilon^3}-1\right)$ gives the correct coefficient before the temperature gradient in the neoclassical electric field in the plateau and banana regime

$$\nu_* = \frac{c(y)}{2\pi\varepsilon^{3/2}\tau_a\sqrt{T_i/m_a}}, \qquad c(y) = \oint h_x |b_x|^{-1} dx, \qquad \varepsilon = \frac{\left(\oint h_x dx\right)^2}{\oint \sqrt{g}/h_y dx}$$

Old form

$$\frac{3}{2}\frac{\partial n_e T_e}{\partial t} + \frac{1}{\sqrt{g}}\frac{\partial}{\partial x}\left(\frac{\sqrt{g}}{h_x}\tilde{q}_{ex}\right) + \frac{1}{\sqrt{g}}\frac{\partial}{\partial y}\left(\frac{\sqrt{g}}{h_y}\tilde{q}_{ey}\right) + \frac{n_e T_e}{\sqrt{g}}\frac{\partial}{\partial x}\left(\frac{\sqrt{g}}{h_x}b_x V_{e\|}\right) =$$
$$= Q_e + c_{E\times B} n_e T_e B_z \frac{1}{h_x h_y}\left(\frac{\partial\phi}{\partial y}\frac{\partial}{\partial x}\left(\frac{1}{B^2}\right) - \frac{\partial\phi}{\partial x}\frac{\partial}{\partial y}\left(\frac{1}{B^2}\right)\right) - \frac{j_y^{(ST)}}{e n_e}\frac{\partial n_e T_e}{h_y \partial y} + j_x \hat{E}_x + 0\cdot j_y \hat{E}_y, \quad (3)$$

New form is used when b2sigp_style is 2

$$\frac{3}{2}\frac{\partial n_e T_e}{\partial t} + \frac{1}{\sqrt{g}}\frac{\partial}{\partial x}\left(\frac{\sqrt{g}}{h_x}\tilde{q}_{ex}\right) + \frac{1}{\sqrt{g}}\frac{\partial}{\partial y}\left(\frac{\sqrt{g}}{h_y}\tilde{q}_{ey}\right) + \frac{n_e T_e}{\sqrt{g}}\frac{\partial}{\partial x}\left(\frac{\sqrt{g}}{h_x}b_x V_{e\|}\right) =$$
$$= Q_e + c_{E\times B} n_e T_e B_z \frac{1}{h_x h_y}\left(\frac{\partial\phi}{\partial y}\frac{\partial}{\partial x}\left(\frac{1}{B^2}\right) - \frac{\partial\phi}{\partial x}\frac{\partial}{\partial y}\left(\frac{1}{B^2}\right)\right) - \frac{j_y^{(ST)}}{e n_e}\frac{\partial n_e T_e}{h_y \partial y} + Q_{F_{ei}}, \quad (3)$$

$$Q_{F_{ei}} = V_{e\|}\sum_{\substack{a=0\\ z_a\neq 0}}^{n_s-1}\left(S^m_{fr,ea} + S^m_{Term,ea}\right)$$

$$Q_e = -Q_\Delta - Q_R$$

$$Q_\Delta = c_{eqp}\Lambda e^{\frac{3}{2}}\frac{\sum_{a=0}^{n_s-1}\frac{z_a^2 m_p}{m_a}n_a}{n_i}\frac{n_e n_i}{T_e^{\frac{3}{2}}}(T_e - T_i), \quad c_{eqp} = 4{,}8\cdot 10^{-15}, \quad \text{where } \Lambda \text{ can be chosen as}$$

$$\Lambda = \begin{cases} 12 \\ \begin{cases} \max\left\{5;\ 23{,}4 - 1{,}15\cdot\log_{10}\left(\frac{n_e}{10^6}\right) + 3{,}45\cdot\log_{10}\left(\frac{T_e}{e}\right)\right\}, & \frac{T_e}{e}\le 50 \\ \max\left\{5;\ 25{,}3 - 1{,}15\cdot\log_{10}\left(\frac{n_e}{10^6}\right) + 2{,}30\cdot\log_{10}\left(\frac{T_e}{e}\right)\right\}, & \frac{T_e}{e} > 50 \end{cases} \quad \text{Braginsky formula} \\ \max\left\{5;\ 15{,}2 - 0{,}5\cdot\log_e\left(\frac{n_e}{10^{20}}\right) + \log_e\left(\frac{T_e}{e\cdot 10^3}\right)\right\} \quad \text{Wesson formula for electron - ion collisions} \end{cases},$$

$Q_R = \sum_{a=0}^{n_s-1} r_{Q_a} n_e n_a$ from b2stel

$$\widetilde{q}_{ex} = \widetilde{f}_{ex} T_e - \kappa_{ex} \frac{1}{h_x}\frac{\partial T_e}{\partial x} + 0 \cdot \alpha_{ex} T_e \hat{E}_x - q_{ex}^{P.Sch}$$

New

$$\widetilde{f}_{ex} = \frac{3}{2}\Gamma_{ex} + V_{ex}^{(Str)} n_e - \widetilde{c}_e^{(2)} \frac{j_x^{(\|)}}{e}$$

Old

$$\widetilde{f}_{ex} = \frac{3}{2}\Gamma_{ex} + V_{ex}^{(Str)} n_e - 0.71 \frac{j_x^{(\|)}}{e}$$

$$q_{ex}^{P.Sch} = -\frac{5}{2} D_{exy} \frac{1}{h_y} \frac{\partial n_e T_e^2}{\partial y}$$

$$D_{exy} = \left(\frac{1}{B^2} - \frac{1}{\left\langle B^2 \right\rangle}\right)\frac{B_z}{e} \qquad D_{eyx} = \left(\frac{1}{B^2} - \frac{1}{\left\langle B^2 \right\rangle}\right)\frac{B_z}{e} \qquad \text{where } \left\langle B^2 \right\rangle = \frac{\iint_{core} B^2 \sqrt{g}\, dxdy}{\iint_{core} \sqrt{g}\, dxdy}$$

New

$$q_{ex} = \left[\frac{3}{2}\Gamma_{ex} - \widetilde{c}_e^{(2)} \frac{j_x^{(\|)}}{e} - \frac{5}{2} n_e T_e \frac{B_z}{e} \frac{\partial}{h_y \partial y}\left(\frac{1}{B^2}\right) + V_{ex}^{(Str)} n_e\right] T_e - \kappa_{ex} \frac{1}{h_x}\frac{\partial T_e}{\partial x} + 0 \cdot \alpha_{ex} T_e \hat{E}_x$$

Old

$$q_{ex} = \left[\frac{3}{2}\Gamma_{ex} - 0.71 \frac{j_x^{(\|)}}{e} - \frac{5}{2} n_e T_e \frac{B_z}{e} \frac{\partial}{h_y \partial y}\left(\frac{1}{B^2}\right) + V_{ex}^{(Str)} n_e\right] T_e - \kappa_{ex} \frac{1}{h_x}\frac{\partial T_e}{\partial x} + 0 \cdot \alpha_{ex} T_e \hat{E}_x$$

$$\Gamma_{ex} = \sum_{a=0}^{n_s - 1} z_a \Gamma_{ax}^{(he)} - \frac{j_x^{(\|)}}{e},$$

$$\Gamma_{ax}^{(he)} = (b_x V_{a\|} + V_{ax}^{(E)} + \frac{5}{3} V_a^{(AN)} + V_a^{corr\_dpc}) n_a - \frac{5}{3}\left[D_a^n \frac{1}{h_x}\frac{\partial n_a}{\partial x} + \widetilde{D}_{ax}^p \frac{1}{h_x}\frac{\partial p_a}{\partial x}\right]$$

New

$$q_{ex}^{(Eir)} = \left[\frac{5}{2}\Gamma_{ex}^{(Eir)} + V_{ex}^{(Str)} n_e - \widetilde{c}_e^{(2)} \frac{\widetilde{J}_\|}{e}\right] T_e - \kappa_{ex} \frac{1}{h_x}\frac{\partial T_e}{\partial x}$$

Old

$$q_{ex}^{(Eir)} = \left[\frac{5}{2}\Gamma_{ex}^{(Eir)} + V_{ex}^{(Str)} n_e - 0.71 \frac{\widetilde{J}_\|}{e}\right] T_e - \kappa_{ex} \frac{1}{h_x}\frac{\partial T_e}{\partial x}$$

$$q_{ey}^{(Eir)} = \left[\frac{5}{2}\Gamma_{ey}^{(Eir)} + V_{ey}^{(Str)} n_e\right] T_e - \kappa_{ey} \frac{1}{h_y}\frac{\partial T_e}{\partial y}$$

where

$$\Gamma_{ex}^{(Eir)} = \left(b_x V_{\|e} + V_{ex}^{(E)} + V_{ex}^{(dia)} + V_{ex}^{(AN)}\right) n_e - D_a^n \frac{1}{h_x}\frac{\partial n_e}{\partial x}$$

$$\Gamma_{ey}^{(Eir)} = \left(V_{ey}^{(E)} + V_{ey}^{(dia)} + V_{ey}^{(AN)}\right) n_e - D_a^n \frac{1}{h_y}\frac{\partial n_e}{\partial y}$$

$$V_{ex}^{(dia)} = \frac{B_z}{eB^2 n_e}\frac{\partial n_e T_e}{h_y \partial y} \qquad V_{ey}^{(dia)} = -\frac{B_z}{eB^2 n_e}\frac{\partial n_e T_e}{h_x \partial x}$$

$$V_{ex}^{(Str)} = 0$$

$$\kappa_{ex} = \kappa_e^{(AN)} + \tilde{\tilde{\kappa}}_{ex}^{(CL)},$$

$$\kappa_e^{(AN)} = \chi_e n_e,$$

$$\chi_e = \left(c_{h_e,0} + c_{h_e,1}\frac{10^{20}}{n_e} + c_{h_e,3}\frac{6{,}2\cdot 10^{-2}T_e}{e|B|}\right) f_{h_e}(x,y),$$

$$c_{h_e,0} = 1.2,\ c_{h_e,1} = 0,\ c_{h_e,3} = 0,$$

$$f_{h_e}(x,y) = \begin{cases} \alpha_{1,h_e}, & -\delta_{1,h_e} \le y \le 0,\ \ x \in Core, \\ \alpha_{2,h_e}, & 0 \le y \le \delta_{2,h_e},\ \ x \in Core, \\ \alpha_{3,h_e}, & x, y \in \Pr \cup SOL \end{cases}$$

$$\tilde{\tilde{\kappa}}_{ex}^{(CL)} = c_{k_{ex}}^{(F\lim)} \tilde{\kappa}_{ex}^{(CL)}$$

$$c_{k_{ex}}^{(F\lim)} = \begin{cases} 1, & \text{only for inner flux surfaces} \\ \dfrac{1}{1 + \dfrac{\frac{\sqrt{g}}{h_x}\tilde{\kappa}_{ex}^{(CL)}\left|\frac{\partial T_e}{h_x \partial x}\right|}{V_{h_e}^{(\max)} n_e T_e}}, & \text{for other flux surfaces} \end{cases}$$

$$V_{h_e}^{(\max)} = c_{f\lim_0}\frac{\sqrt{g}}{h_x}|b_x|\sqrt{\frac{T_e}{m_e}},$$

$$\tilde{\kappa}_{ex}^{(CL)} = c_{\kappa_{ex}^{(CL)}}^{(Luciani)} \kappa_{ex}^{(CL)} b_x^2,$$

For plateau and banana correction ‘`b2trcl_lluciani`’ ‘3’

$$c_{\kappa_{ex}^{(CL)}}^{(Luciani)} = \begin{cases} 0.58\varepsilon^{3/2}/K_1, & \text{only for inner flux surfaces} \\ 1, & \text{for other flux surfaces} \end{cases}$$

$$K_1 = \frac{1}{1 + 2.01\cdot \nu_{*e}^{1/2} + 1.53\cdot \nu_{*e}} + \frac{0.52\cdot\varepsilon^3\cdot\nu_{*e}}{1 + 0.89\cdot\varepsilon^{3/2}\cdot\nu_{*e}}$$

$$\nu_{*e} = \frac{c(y)}{2\pi\varepsilon^{3/2}\tau_e\sqrt{T_e / m_e}} \ ,$$

$$\tau_e = \frac{3}{4\sqrt{2\pi}} \frac{\sqrt{m_e} T_e^{\frac{3}{2}}}{z_{eff} n_e \Lambda} \left(\frac{4\pi\varepsilon_0}{e^2}\right)^2$$, where Coulomb logarithm Λ can be chosen as

$$\Lambda = \begin{cases} 12 \\ \begin{cases} \max\left\{5;\ 23{,}4 - 1{,}15 \cdot \log_{10}\left(\frac{n_e}{10^6}\right) + 3{,}45 \cdot \log_{10}\left(\frac{T_e}{e}\right)\right\}, & \frac{T_e}{e} \le 50 \\ \max\left\{5;\ 25{,}3 - 1{,}15 \cdot \log_{10}\left(\frac{n_e}{10^6}\right) + 2{,}30 \cdot \log_{10}\left(\frac{T_e}{e}\right)\right\}, & \frac{T_e}{e} > 50 \end{cases} & \text{Braginsky formula} \\ \max\left\{5;\ 14{,}9 - 0{,}5 \cdot \log_e\left(\frac{n_e}{10^{20}}\right) + \log_e\left(\frac{T_e}{e \cdot 10^3}\right)\right\} & \text{Wesson formula for electron - electron collisions} \end{cases} ,$$

but it was as below

$$c_{\kappa_{ex}^{(CL)}}^{(Luciani)} = \begin{cases} \dfrac{1}{1 + \dfrac{7{,}5 \cdot 10^{16} \left(\frac{T_e}{e}\right)^2}{c(y) \sum_{a=0}^{n_s - 1} z_a^2 n_a}}, & \text{only for inner flux surfaces} \\ 1, & \text{for other flux surfaces} \end{cases}$$

$$c(y) = \begin{cases} \int \frac{h_x(x,y)}{|b_x|} dx, & \textit{for inner flux surfaces} \\ 0, & \textit{for other} \end{cases}$$

$$\kappa_{ex}^{(CL)} = \frac{5}{2} \frac{T_e}{m_e} f_{k_e}(z_{eff}) n_e \tau_e$$, old form

$$f_{k_e}(z_{eff}) = \frac{2{,}16 z_{eff}}{1 + 0{,}27 z_{eff}}$$, old form

$$\kappa_{ex}^{(CL)} = \frac{T_e}{m_e} f_{k_e}(z_{eff}) n_e \tau_e$$, new form

$$f_{k_e}(z_{eff}) = \frac{3{,}9 + 2{,}3 / z_{eff}}{0{,}31 + 1{,}2 / z_{eff} + 0{,}41 / z_{eff}^2}$$, new Zhdanov's expression

$$z_{eff} = \frac{\sum_{a=0}^{n_s - 1} z_a^2 n_a}{n_e} ,$$

$$V_{e\|} = \frac{1}{en_e}\left[\sum_{a=0}^{n_s-1} z_a e n_a V_{a\|} - \frac{j_x^{(\|)}}{b_x}\right],$$

New expression of parallel current

$$j_x^{(\|)} = \sigma_\| \left(\frac{1}{en_e h_x}\frac{\partial n_e T_e}{\partial x} - \frac{1}{h_x}\frac{\partial \varphi}{\partial x}\right) - \alpha_{ex}\frac{1}{h_x}\frac{\partial T_e}{\partial x} - \frac{eb_x}{z_{eff}}\sum_{a=0}^{n_s-1} n_a V_{a\|}\left(z_a^2 - z_a z_{eff}\right)$$

Old

$$j_x^{(\|)} = \sigma_\| \hat{E}_x - \alpha_{ex}\frac{1}{h_x}\frac{\partial T_e}{\partial x},$$

$$\hat{E}_x = \frac{1}{en_e h_x}\frac{\partial n_e T_e}{\partial x} - \frac{1}{h_x}\frac{\partial \phi}{\partial x},$$

$$\sigma_\| = \sigma_{AN\|} + \tilde{\sigma}_{CL\|},$$

$$\sigma_{AN\|} = 0$$

$$\tilde{\sigma}_{CL\|} = c_{\sigma_{CL\|}}^{(Luciani)} \sigma_{CL\|}$$

For plateau and banana correction '`b2trcl_lluciani`' '3'

$$c_{\sigma_{CL\|}}^{(Luciani)} = \begin{cases} 0.58\varepsilon^{3/2}/K_1, & \text{only for inner flux surfaces} \\ 1, & \text{for other flux surfaces} \end{cases}$$

Old form

$$c_{\sigma_{CL\|}}^{(Luciani)} = \frac{1}{1 + \dfrac{7{,}5\cdot 10^{16}\left(\dfrac{T_e}{e}\right)^2}{c(y)\sum_{a=0}^{n_s-1} z_a^2 n_a}}$$

New form

$$\sigma_{CL\|} = b_x^2 \frac{e^2}{m_e} f_{\sigma_e}\left(z_{eff}\right)\tau_e n_e z_{eff}, \qquad f_{\sigma_e}\left(z_{eff}\right) = \frac{1}{c_e^{(1)} z_{eff}}$$

Old form

$$\sigma_{CL\|} = b_x^2 \frac{e^2}{m_e} f_{\sigma_e}\left(z_{eff}\right) n_e \tau_e, \qquad f_{\sigma_e}\left(z_{eff}\right) = \frac{4{,}95 z_{eff}}{1 + 1{,}54 z_{eff}},$$

$$\alpha_{ex} = \alpha_{ANx} + \tilde{\tilde{\alpha}}_{CLx},$$

$$\alpha_{ANx} = 0,$$

$$\tilde{\tilde{\alpha}}_{CLx} = c_{k_e}^{(F\lim)} \tilde{\alpha}_{CLx}$$

$$\tilde{\alpha}_{CLx} = c^{(Luciani)}_{k^{(CL)}{}_{ex}} \alpha_{CLx}$$

New form

$$\tilde{\tilde{\alpha}}_{CLx} = c^{(F\lim)}_{\alpha_{ex}} \tilde{\alpha}_{CLx}, \quad \tilde{\alpha}_{CLx} = c^{(Luciani)}_{\alpha^{(CL)}_{ex}} \alpha_{CLx}, \quad \alpha_{CLx} = b_x^2 \frac{e}{m_e} f_\alpha\left(z_{eff}\right)\tau_e n_e z_{eff}, \quad f_\alpha\left(z_{eff}\right) = -\frac{\tilde{c}^{(2)}_e}{c^{(1)}_e}$$

$$c^{(F\lim)}_{\alpha_{ex}} = \begin{cases} 1, & \text{inside the separatrix} \\ \dfrac{1}{1+\left|\dfrac{\alpha_{CLx}\dfrac{1}{h_x}\dfrac{\partial T_e}{\partial x}}{c_{\text{flime}} j_{e\max}}\right|}, & \text{outside the separatrix} \end{cases} \qquad j_{e\max} = \left|b_x\right| e n_e \sqrt{\frac{T_e}{m_e}}$$

New coefficient c_{flime} is stored in cflim(3)

Old form

$$\alpha_{CLx} = b_x^2 \sqrt{\frac{5}{2}\frac{e}{m_e}} f_\alpha\left(z_{eff}\right) n_e \tau_e, \qquad f_\alpha\left(z_{eff}\right) = -\frac{1,25 z_{eff}}{1+0,41 z_{eff}},$$

$$\tilde{q}_{ey} = \tilde{f}_{ey} T_e - \kappa_{ey}\frac{1}{h_y}\frac{\partial T_e}{\partial y} + 0\cdot\alpha_{ey} T_e \hat{E}_y - q^{P.Sch}_{ey}$$

$$\tilde{f}_{ey} = \frac{3}{2}\left(\Gamma_{ey} - \frac{5}{3}\frac{j^{(ST)}_y}{e}\right) + V^{(Str)}_{ey} n_e$$

$$q^{P.Sch}_{ey} = \frac{5}{2} D_{eyx}\frac{1}{h_x}\frac{\partial n_e T_e^2}{\partial x}$$

$$q_{ey} = \left[\frac{3}{2}\Gamma_{ey} - \frac{5}{2}\frac{j^{(ST)}_y}{e} + \frac{5}{2} n_e T_e \frac{B_z}{e}\frac{\partial}{h_x \partial x}\left(\frac{1}{B^2}\right) + V^{(Str)}_{ey} n_e\right] T_e - \kappa_{ey}\frac{1}{h_y}\frac{\partial T_e}{\partial y} + 0\cdot\alpha_{ey} T_e \hat{E}_y$$

$$\Gamma_{ey} = \sum_{a=0}^{n_s-1} z_a \Gamma^{(he)}_{ay}$$

$$\Gamma^{(he)}_{ay} = \left(\frac{5}{3}V^{(AN)}_a + V^{(E)}_{ay}\right) n_a - \frac{5}{3}\left[D^n_a \frac{1}{h_y}\frac{\partial n_a}{\partial y} + \tilde{D}^p_{ay}\frac{1}{h_y}\frac{\partial p_a}{\partial y}\right]$$

$$V^{(Str)}_{ey} = 0$$

$$\hat{E}_y = \frac{1}{e n_e h_y}\frac{\partial n_e T_e}{\partial y} - \frac{1}{h_y}\frac{\partial \phi}{\partial y},$$

$$\kappa_{ey} = \kappa^{(AN)}_e + \kappa^{(CL)}_{ey} + \kappa^{(ST)}_e,$$

$$\kappa^{(CL)}_{ey} = 0$$

$$\kappa_e^{(ST)} = \begin{cases} 1.6\alpha^{(ST)}\sigma^{(ST)}T_e/e^2, & y \in \left[-\Delta^{(ST)},0\right] \\ 0, & y \notin \left[-\Delta^{(ST)},0\right] \end{cases}, \qquad \alpha^{(ST)} \approx 2$$

$$\alpha_{ey} = \alpha_{ANy} + \alpha_{CLy},$$

$$\alpha_{ANy} = \left(c_{\alpha,0} + c_{\alpha,1}\frac{10^{20}}{n_e} + c_{\alpha,3}\frac{6{,}2\cdot 10^{-2}T_e}{e|B|}\right)n_e\sqrt{\frac{e}{T_e}}$$

$$c_{\alpha,0} = 10^{-3},\ c_{\alpha,1} = 0,\ c_{\alpha,3} = 0$$

$$\alpha_{CLy} = 0$$

$$\frac{3}{2}\frac{\partial n_i T_i}{\partial t} + \frac{1}{\sqrt{g}}\frac{\partial}{\partial x}\left(\frac{\sqrt{g}}{h_x}\tilde{q}_{ix}\right) + \frac{1}{\sqrt{g}}\frac{\partial}{\partial y}\left(\frac{\sqrt{g}}{h_y}\tilde{q}_{iy}\right) + \sum_{a=0}^{n_s-1}\frac{n_a T_i}{\sqrt{g}}\frac{\partial}{\partial x}\left(\frac{\sqrt{g}}{h_x}V_{a\|}b_x\right) =$$

$$= Q_\Delta + Q_{F_{ab}} + c_{E\times B}T_i B_z\frac{1}{h_x h_y}\left(\frac{\partial\phi}{\partial y}\frac{\partial}{\partial x}\left(\frac{1}{B^2}\right) - \frac{\partial\phi}{\partial x}\frac{\partial}{\partial y}\left(\frac{1}{B^2}\right)\right)\sum_{\substack{a=0\\ z_a\neq 0}}^{n_s-1}n_a +$$

$$+\sum_{a=0}^{n_s-1}\left(\eta_{ax}\left(\frac{\partial V_{a\|}}{h_x\partial x}\right)^2 + \eta_{ANa}\left(\frac{\partial V_{a\|}}{h_y\partial y}\right)^2\right) + Q_I^{(i)} + Q_R^{(i)},$$

If Eirene is working then: $\sum_{\substack{a=0\\ z_a\neq 0}}^{n_s-1}\left(\eta_{ax}\left(\frac{\partial V_{a\|}}{h_x\partial x}\right)^2 + \eta_{ANa}\left(\frac{\partial V_{a\|}}{h_y\partial y}\right)^2\right)$

Contribution from friction between charged species

Old form

$$Q_{F_{ab}} = \sum_{\substack{b=0\\ z_b\neq 0}}^{n_s-1}\sum_{\substack{a=b+1\\ z_a\neq 0}}^{n_s-1}\frac{1}{4\tau_p}m_p\sqrt{\frac{\frac{m_a}{m_p}\frac{m_b}{m_p}}{\frac{m_a}{m_p}+\frac{m_b}{m_p}}}z_a^2 z_b^2 n_a n_b\left(V_{a\|} - V_{b\|}\right)^2$$

New form is used when b2sigp_style is 2

$$Q_{F_{ab}} = -\sum_{\substack{a=0\\ z_a\neq 0}}^{n_s-1}V_{a\|}\left(S_{fr,a}^m + S_{Term,a}^m\right)$$

Atom heat source due to ionization

$$Q_I^{(i)} = \sum_{a=0}^{n_s-2}Q_{I_a}^{(i)},$$

$$Q_{I_a}^{(i)} = \begin{cases} r_{I_a}n_e n_a\left(\frac{m_a + m_{a+1}}{4}\right)\left(V_{\|a} - V_{\|a+1}\right)^2, & z_a < z_{a+1}\ \ \&\ \ z_n = z_{n+1} \\ 0 \end{cases}$$

Atom heat source due to recombination

$$Q_R^{(i)} = \sum_{a=1}^{n_s-1} Q_{R_a}^{(i)},$$

$$Q_{R_a}^{(i)} = \begin{cases} r_{R_a} n_e n_a \left(\dfrac{m_a + m_{a-1}}{4}\right)\left(V_{\|a} - V_{\|a-1}\right)^2, & z_{a-1} < z_a \ \& \ z_{n-1} = z_n \\ 0 \end{cases}$$

$\tau_p = \dfrac{3}{4\sqrt{2\pi}} \dfrac{\sqrt{m_p} T_i^{\frac{3}{2}}}{\Lambda} \left(\dfrac{4\pi\varepsilon_0}{e^2}\right)^2$, where the Coulomb logarithm Λ can be chosen as

$$\Lambda = \begin{cases} 12 \\ \begin{cases} 23{,}4 - 1{,}15 \cdot \log_{10}\left(\dfrac{n_e}{10^6}\right) + 3{,}45 \cdot \log_{10}\left(\dfrac{T_e}{e}\right), & \dfrac{T_e}{e} \le 50 \\ 25{,}3 - 1{,}15 \cdot \log_{10}\left(\dfrac{n_e}{10^6}\right) + 2{,}30 \cdot \log_{10}\left(\dfrac{T_e}{e}\right), & \dfrac{T_e}{e} > 50 \end{cases} & \text{Braginsky formula} \\ \max\left\{5;\ 17{,}3 - 0{,}5 \cdot \log_e\left(\dfrac{n_e}{10^{20}}\right) + 1{,}5 \cdot \log_e\left(\dfrac{T_e}{e \cdot 10^3}\right)\right\} & \text{Wesson formula for ion-ion collisions} \end{cases},$$

$$\tilde{q}_{ix} = \tilde{f}_{ix} T_i - \kappa_{ix} \frac{1}{h_x} \frac{\partial T_i}{\partial x} - q_{ix}^{P.Sch}$$

$$\tilde{f}_{ix} = \frac{3}{2}\Gamma_{ix} + V_{ix}^{(Str)} n_i + \frac{3}{2} \frac{j_x^{(AN)} + j_x^{(in)} + \tilde{j}_x^{(vis\|)} + \tilde{j}_x^{(visq)} + j_x^{(vis\perp)}}{e}$$

$$q_{ix}^{P.Sch} = \frac{5}{2} \sum_{\substack{a=0 \\ z_a \ne 0}}^{n_s-1} D_{axy} \frac{1}{h_y} \frac{\partial n_a T_i^2}{\partial y}$$

$$q_{ix} = \left[\frac{3}{2}\Gamma_{ix} + \frac{5}{2} \sum_{\substack{a=0 \\ z_a \ne 0}}^{n_s-1} n_a \tilde{V}_{ax}^{(dia)} + V_{ix}^{(Str)} n_i + \frac{3}{2} \frac{j_x^{(AN)} + j_x^{(in)} + \tilde{j}_x^{(vis\|)} + \tilde{j}_x^{(visq)} + j_x^{(vis\perp)}}{e}\right] T_i - \kappa_{ix} \frac{1}{h_x} \frac{\partial T_i}{\partial x}$$

$$\Gamma_{ix} = \sum_{a=0}^{n_s-1} \Gamma_{ax}^{(he)}$$

$$q_{ix}^{(Eir)} = \frac{5}{2} \sum_{\substack{a=0 \\ z_a \ne 0}}^{n_s-1} \Gamma_{ax}^{(Eir)} T_i - \kappa_{ix} \frac{1}{h_x} \frac{\partial T_i}{\partial x}$$

$$\kappa_{ix} = \kappa_{ix}^{(AN)} + \approx{\kappa}_{ix}^{(CL)},$$

$$k_{ix}^{(AN)} = \sum_{a=0}^{n_s-1} \tilde{k}_{ax}^{(AN)},$$

$$\tilde{k}_{ax}^{(AN)} = \begin{cases} k_a^{(AN)}, & z_a \neq 0 \\ \dfrac{k_a^{(AN)}}{\left(1+\left(\dfrac{\dfrac{\sqrt{g}}{h_x} k_a^{(AN)} \left|\dfrac{\partial T_i}{h_x \partial x}\right|}{\dfrac{\sqrt{g}}{h_x} \alpha V_{h_i} n_a T_i}\right)^{\gamma_2}\right)^{\frac{1}{\gamma_2}}}, & z_a = 0 \end{cases}$$

$$V_{h_i} = \sqrt{\frac{2T_i}{m_a}}$$

$$k_a^{(AN)} = \chi_a^{(AN)} n_a,$$

$$\chi_a^{(AN)} = \left(c_{h_a,0} + c_{h_a,1}\frac{10^{20}}{n_e} + c_{h_a,2} D_a^f + c_{h_a,3}\frac{6{,}2\cdot 10^{-2} T_e}{e|B|}\right) f_{h_a}(x,y),$$

$$a=0,\ c_{h_a,0}=0,\ c_{h_a,1}=0,\ c_{h_a,2}=1.4,\ c_{h_a,3}=0,$$

$$a=1,\ c_{h_a,0}=0,\ c_{h_a,1}=0,\ c_{h_a,2}=1.2,\ c_{h_a,3}=0,$$

$$f_{h_a}(x,y) = \begin{cases} 1, & z_a = 0 \\ \alpha_{1,h_a}, & -\delta_{1,h_a} \le y \le 0,\ x \in Core,\ z_a \neq 0 \\ \alpha_{2,h_a}, & 0 \le y \le \delta_{2,h_a},\ x \in Core, \\ \alpha_{3,h_a}, & x, y \in \Pr \cup SOL \end{cases}$$

$$\tilde{\tilde{k}}_{ix}^{(CL)} = c_{k_{ix}}^{(F\lim)} \tilde{k}_{ix}^{(CL)}$$

$$c_{k_{ix}}^{(F\lim)} = \begin{cases} 1, & \textit{for inner flux surfaces} \\ \dfrac{1}{1+\dfrac{\dfrac{\sqrt{g}}{h_x} k_{ix}^{(CL)} \left|\dfrac{\partial T_i}{h_x \partial x}\right|}{V_{h_i}^{(\max)} n_i T_i}}, & \textit{for other flux surfaces} \end{cases},$$

$$V_{h_i}^{(\max)} = c_{f\lim_1} \frac{\sqrt{g}}{h_x} |b_x| \sqrt{\frac{T_i}{m_p}} \frac{\sum_{a=0}^{n_s-1} n_a \sqrt{\dfrac{m_p}{m_a}}}{n_i},$$

$$c_{f\lim_1} = 0{,}15$$

$$\tilde{\kappa}_{ix}^{(CL)} = c_{k_{ix}}^{(Luciani)} \kappa_{ix}^{(CL)},$$

‘`b2trcl_lluciani`’ ‘1’

$$c_{k_{ix}}^{(Luciani)} = \begin{cases} \dfrac{1}{1+\dfrac{7{,}5\cdot 10^{16}\left(\dfrac{T_i}{e}\right)^2}{c(y)\sum\limits_{a=0}^{n_s-1} z_a^2 n_a}}, & \textit{for inner flux surfaces} \\ 1, & \textit{for other flux surfaces} \end{cases}$$

For plateau and banana correction ‘`b2trcl_lluciani`’ ‘3’

$$c_{k_{ix}}^{(Luciani)} = 1.6\cdot\varepsilon^{3/2}/K_2,$$

$$K_2 = \frac{0.66+1.88\cdot\sqrt{\varepsilon}-1.54\cdot\varepsilon}{1+1.03\cdot\nu_*^{1/2}+0.31\cdot\nu_*} + \frac{1.17\cdot\varepsilon^3\cdot\nu_*}{1+0.74\cdot\varepsilon^{3/2}\cdot\nu_*}$$

$$\kappa_{ix}^{(CL)} = \frac{5}{2} b_x^2 \sum_{\substack{a=0\\ z_a\neq 0}}^{n_s-1} \frac{T_i}{m_a} n_a f_{k_i}\tau_a,$$

$$f_{k_i} = 2.253$$

For the integral modeling ('b2trcl_integral_modeling' ‘1’) in the region of the overlap with ASTRA in the inner part of the core:

$$\tilde{\kappa}_{ix}^{(CL)} = \frac{25}{4}\left(n_1 T_i \frac{B_z}{B_x h_y}\right)^2 \frac{h_{y(outer\,midplane)}}{e}\left[\oint \frac{\sqrt{g}}{B^2}\left(1-\frac{B^2}{\langle B^2\rangle}\right)^2 dx\right]\times K_{ASTRA},$$

here $\langle B^2\rangle = \dfrac{\oint B^2\sqrt{g}\,dx}{\oint\sqrt{g}\,dx}$

from ASTRA we get T_i/e and $\oint q_i^{NEO} dS$ as a functions of $R_{(outer\,midplane)}$ and calculate

$K_{ASTRA} = \dfrac{-1}{\oint q_i^{NEO} dS}\dfrac{d(T_i/e)}{dR_{(outer\,midplane)}}$ as a function of the flux surface in B2

$$V_{ix}^{(Str)} = 0$$

$$\tilde{q}_{iy} = \tilde{f}_{iy} T_i - \kappa_{iy}\frac{1}{h_y}\frac{\partial T_i}{\partial y} - q_{iy}^{P.Sch}$$

$$\tilde{f}_{iy} = \frac{3}{2}\Gamma_{iy} + V_{iy}^{(Str)} n_i + \frac{3}{2}\frac{\left(j_y^{(AN)} + j_y^{(in)} + \tilde{j}_y^{(vis\|)} + \tilde{j}_y^{(visq)} + j_y^{(vis\perp)}\right)}{e}$$

$$q_{iy}^{P.Sch} = -\frac{5}{2}\sum_{\substack{a=0\\ z_a\neq 0}}^{n_s-1} D_{ayx}\frac{1}{h_x}\frac{\partial n_a T_i^2}{\partial x}$$

$$q_{iy}=\left[\frac{3}{2}\Gamma_{iy}+\frac{5}{2}\sum_{\substack{a=0\\z_a\neq 0}}^{n_s-1}n_a\tilde{V}_{ay}^{(dia)}+V_{iy}^{(Str)}n_i+\frac{3}{2}\beta_{\mathrm{Scott}}\frac{j_y^{(in)}}{e}+\frac{3}{2}\frac{\left(j_y^{(AN)}+j_y^{(in)}+\tilde{j}_y^{(vis\|)}+\tilde{j}_y^{(visq)}+j_y^{(vis\perp)}\right)}{e}\right]T_i-\kappa_{iy}\frac{1}{h_y}\frac{\partial T_i}{\partial y}$$

$$\beta_{\mathrm{Scott}}=\begin{cases}0, & for\ us\\ 1, & for\ Bruce\ \mathrm{Scott}\end{cases}$$

$$\Gamma_{iy}=\sum_{a=0}^{n_s-1}\Gamma_{ay}^{(he)}$$

$$q_{iy}^{(Eir)}=\frac{5}{2}\sum_{\substack{a=0\\z_a\neq 0}}^{n_s-1}\Gamma_{ay}^{(Eir)}T_i-\kappa_{iy}\frac{1}{h_y}\frac{\partial T_i}{\partial y}$$

$$V_{iy}^{(Str)}=0$$

$$\kappa_{iy}=\kappa_{iy}^{(AN)}+\kappa_{iy}^{(CL)},$$

$$k_{iy}^{(AN)}=\sum_{a=0}^{n_s-1}\tilde{k}_{ay}^{(AN)},$$

$$\tilde{k}_{ay}^{(AN)}=\begin{cases}k_a^{(AN)}, & z_a\neq 0\\ \dfrac{k_a^{(AN)}}{\left(1+\left(\dfrac{\dfrac{\sqrt{g}}{h_y}k_a^{(AN)}\left|\dfrac{\partial T_i}{h_y\partial y}\right|}{\dfrac{\sqrt{g}}{h_y}\alpha V_{h_i}n_aT_i}\right)^{\gamma_2}\right)^{\frac{1}{\gamma_2}}}, & z_a=0\end{cases}$$

$$\kappa_{iy}^{(CL)}=0$$

$$\frac{1}{\sqrt{g}}\frac{\partial}{\partial x}\left(\frac{\sqrt{g}}{h_x}j_x\right)+\frac{1}{\sqrt{g}}\frac{\partial}{\partial y}\left(\frac{\sqrt{g}}{h_y}j_y\right)=0$$

$$j_x=j_x^{(AN)}+\tilde{j}_x^{(dia)}+j_x^{(in)}+\tilde{j}_x^{(vis\|)}+\tilde{j}_x^{(vis\perp)}+\tilde{j}_x^{(visq)}+\tilde{j}_x^{(s)}+j_x^{(\|)},$$

$$j_x^{(AN)}=-\sigma_{AN}\frac{\partial\phi}{h_x\partial x}$$

$$\tilde{j}_x^{(dia)}=B_z\frac{\sum\limits_{\substack{a=0\\z_a\neq 0}}^{n_s-1}n_a\left(z_aT_e+T_i\right)}{h_y}\frac{\partial}{\partial y}\left(\frac{1}{B^2}\right),$$

$$j_x^{(in)} = \frac{B_z}{2}\frac{\partial}{h_y\partial y}\left(\frac{1}{B^2}\right)\sum_{\substack{a=0\\ z_a\neq 0}}^{n_s-1} m_a n_a V_{a\|}^2 ,$$

$$\tilde{j}_x^{(vis\|)} = -\frac{B_z\sqrt{B}}{3B_x}\frac{\partial}{h_y\partial y}\left(\frac{1}{B^2}\right)\sum_{\substack{a=0\\ z_{n_a}=1\ \&\ z_a=1}}^{n_s-1} \tilde{\tilde{\eta}}_{CLax}\frac{\partial\left(\sqrt{B}V_{a\|}\right)}{h_x\partial x}, \qquad (\tilde{\tilde{\eta}}_{CLax}\ \text{has}\ b_x^2)$$

$$\tilde{j}_x^{(vis\perp)} = 0 ,$$

$$\tilde{j}_x^{(visq)} = -0.24\sum_{\substack{a=0\\ z_{n_a}=1\,\&\,z_a=1}}^{n_s-1} \alpha^{NEO}\tilde{\tau}_a b_z\frac{B_x}{B^{\frac{1}{2}}}\frac{\partial}{h_y\partial y}\left(\frac{1}{B^2}\right)\frac{\partial\left(\tilde{q}_a^{(0)}B^{\frac{1}{2}}\right)}{h_x\partial x}, \quad \tilde{q}_a^{(0)}\ \text{see above}$$

$$\tilde{j}_x^{(s)} = \sum_{\substack{a_0=0\\ z_{a_0}=0\ \&\ z_{n_{a0}}=1}}^{n_s-1}\ \sum_{\substack{a=0\\ z_a\neq 0\ \&\ z_{n_a}=1}}^{n_s-1}\left(-\sigma_{aa_0}b_z^2\frac{\partial\phi}{h_x\partial x} - \sigma_{aa_0}b_z^{\,2}\frac{1}{Z_a e n_a}\frac{\partial n_a T_i}{h_x\partial x} + \sigma_{aa_0}B_z V_{a_0 y}\right),$$

$$V_{a_0 y} = \frac{\Gamma_{a_0 y}}{n_{a_0}}$$

$$\sigma_{aa_0} = \begin{cases}\dfrac{m_a n_a\left\langle V_{aa_0}\sigma_{ex}\right\rangle n_{a_0}}{2B^2}, & z_{a_0}=0\ \&\ z_{n_{a0}}=1,\ z_a\neq 0\ \&\ z_{n_a}=1\\ 0 & \end{cases},$$

$$\left\langle V_{aa_0}\sigma_{ex}\right\rangle = 3.2\cdot 10^{-15}\sqrt{\frac{T_i}{0.026e}}$$

$j_x^{(\|)}$ - see above

$$j_y = j_y^{(AN)} + \tilde{j}_y^{(dia)} + j_y^{(in)} + \tilde{j}_y^{(vis\|)} + j_y^{(vis\perp)} + \tilde{j}_y^{(visq)} + \tilde{j}_y^{(s)} + j_y^{(ST)} ,$$

$$j_y^{(AN)} = -\sigma_{AN}\frac{\partial\phi}{h_y\partial y},$$

$$\tilde{j}_y^{(dia)} = -\frac{B_z\sum\limits_{\substack{a=0\\ z_a\neq 0}}^{n_s-1} n_a\left(z_a T_e + T_i\right)}{h_x}\frac{\partial}{\partial x}\left(\frac{1}{B^2}\right),$$

$$j_y^{(in)} = -\frac{B_z}{2}\frac{\partial}{h_x\partial x}\left(\frac{1}{B^2}\right)\sum_{\substack{a=0\\ z_a\neq 0}}^{n_s-1} m_a n_a V_{a\|}^2 ,$$

$$\tilde{j}_y^{(vis\|)} = \frac{B_z\sqrt{B}}{3B_x}\frac{\partial}{h_x\partial x}\left(\frac{1}{B^2}\right)\sum_{\substack{a=0\\ z_{n_a}=1\ \&\ z_a=1}}^{n_s-1}\tilde{\tilde{\eta}}_{CLax}\frac{\partial\left(B^{\frac{1}{2}}V_{a\|}\right)}{h_x\partial x}, \qquad (\tilde{\tilde{\eta}}_{CLax}\ \text{has}\ b_x^2)$$

$$j_y^{(vis\perp)} = -\sum_{\substack{a=0\\ z_{n_a}=1\ \&\ z_a=1}}^{n_s-1}\frac{1}{B\sqrt{g}}\frac{\partial}{\partial y}\left(\frac{\sqrt{g}}{h_y}\frac{\eta_{ANa}}{4}\frac{\partial V_{a\perp}}{h_y\partial y}\right),$$

$$V_{a\perp} = -\frac{1}{Bh_y}\frac{\partial\phi}{\partial y} - \frac{1}{z_a eBh_y n_a}\frac{\partial n_a T_i}{\partial y}$$

$$\tilde{j}_y^{(visq)} = 0.24\sum_{\substack{a=0\\ z_{n_a}=1\,\&\,z_a=1}}^{n_s-1}\alpha^{NEO}\tilde{\tau}_a b_z\frac{B_x}{B^{\frac{1}{2}}}\frac{\partial}{h_x\partial x}\left(\frac{1}{B^2}\right)\frac{\partial\left(\tilde{q}_a^{(0)}B^{\frac{1}{2}}\right)}{h_x\partial x}, \quad \tilde{q}_a^{(0)}\ \text{see above}$$

$$\tilde{j}_y^{(s)} = \sum_{\substack{a_0=0\\ z_{a_0}=0\ \&\ z_{n_{a_0}}=1}}^{n_s-1}\ \sum_{\substack{a=0\\ z_a\neq 0\ \&\ z_{n_a}=1}}^{n_s-1}\left(-\sigma_{aa_0}\frac{\partial\phi}{h_y\partial y} - \sigma_{aa_0}\frac{1}{z_a en_a}\frac{\partial n_a T_i}{h_y\partial y} - \sigma_{aa_0}B_z V_{a_0 x}\right),$$

$$V_{a_0 x} = \frac{\Gamma_{a_0 x}}{n_{a_0}}$$

σ_{aa_0} see above

$$j_y^{(ST)} = \begin{cases} -\sigma^{(ST)}\left(\frac{1}{h_y}\frac{\partial\phi}{\partial y} - \frac{T_e}{en_e}\frac{1}{h_y}\frac{\partial n_e}{\partial y} - \frac{0.5}{e}\frac{1}{h_y}\frac{\partial T_e}{\partial y}\right), & y\in\left[-\Delta^{(ST)},0\right] \\ 0,\ y\notin\left[-\Delta^{(ST)},0\right] \end{cases}$$

$$\sigma^{(ST)} = \begin{cases} c_{\sigma^{(ST)}}en_e, & y\in\left[-\Delta^{(ST)},0\right] \\ 0, & y\notin\left[-\Delta^{(ST)},0\right] \end{cases}, \qquad c_{\sigma^{(ST)}} > c_{\sigma_{ANy}}$$

$$\sigma_y = \sigma_{AN} + \sigma_{CLy}$$

σ_{AN} depends on 'b2tqna_model_sig'

$$\sigma_{AN} = \left(c_{\sigma_a,0} + c_{\sigma_a,1}\frac{10^{20}}{n_e} + c_{\sigma_a,3}\frac{6{,}2\cdot 10^{-2}T_e}{e|B|} + c_{\sigma_a,4}e^{-c_{\sigma_a,5}\left(\frac{y}{L(x)}\right)^2}\right)en_e,$$

or

$$\sigma_{AN} = \left(c_{\sigma_a,0} + c_{\sigma_a,1}\frac{10^{20}}{n_e} + c_{\sigma_a,3}\frac{6{,}2\cdot 10^{-2}T_e}{e|B|} + c_{\sigma_a,4}e^{-c_{\sigma_a,5}\left(\frac{y}{L(x)}\right)^2}\right)e\tilde{n}_e,\ \tilde{n}_e - density\ at\ inner\ flux\ surface$$

$$L(x) = \int_0^{y_{out}} h_y(x,y)dy$$

$c_{\sigma_a,0} = 10^{-6} - 10^{-3}$, $c_{\sigma_a,1} = 0$, $c_{\sigma_a,3} = 0$,

$\sigma_{CLy} = 0$

Typical boundary conditions

i. Particle balance equation

$$S_{n_a} = S_{n_a,0} + S_{n_a,1} n_a$$

South. Private region

$z_a = 0$

$$\left.\frac{\sqrt{g}}{h_y}\Gamma_{ay}\right|_{S_0} = \left[\left. p_{n_a,a,1}\frac{\sqrt{g}}{h_y}\sqrt{\frac{z_a T_e + T_i}{m_a}} n_a\right|_{S_0} + \sum_{a=0}^{n_s-1} r_{rec,a,0} c_{nsr} \max\left\{0, -\frac{\sqrt{g}}{h_y}\Gamma_{ay}\right\}\right]\right|_{S_0}$$ leakage, **BCCON=10**

$p_{n_a,a,1} = -10^{-2}$, $r_{rec,0,0} = 0$, $r_{rec,1,0} = 1$

$\tilde{\Gamma}_{ay}$ was replaced by Γ_{ay}

c_{nsr} -neutral sources rescale coefficient. 1 – when working B2 and 10^{-10} when eirene coupling

$z_a \neq 0$

BCCON=9 is below

$$\left.\frac{\sqrt{g}}{h_y}\tilde{\Gamma}_{ay}\right|_{S_0} = \begin{cases} \left.\left[-\frac{1}{p_{n_a,a,1}}\frac{\sqrt{g}}{h_y}\left(D^n_{ANa} + D^P_{ANa}(z_a T_e + T_i)\right) n_a\right]\right|_{S_0}, & mdf\_fnb = 0 \\ \left.\left[-\frac{1}{p_{n_a,a,1}}\frac{\sqrt{g}}{h_y}\left(D^n_{ANa} + D^P_{ANa}(z_a T_e + T_i)\right) n_a + \frac{\sqrt{g}}{h_y} D_{ayx}\frac{1}{h_x}\frac{\partial n_a T_i}{\partial x} + \frac{\sqrt{g}}{h_y} V^{(E)}_{ay} n_a\right]\right|_{S_0}, & mdf\_fnb = 1 \end{cases}$$

$$S^{no\_mdf}_{n_a,0} = 0$$

$$S^{no\_mdf}_{n_a,1} = -\frac{1}{p_{n_a,a,1}}\frac{\sqrt{g}}{h_y}\left(D^n_{ANa} + D^P_{ANa}(z_a T_e + T_i)\right)$$

BCCON=10 (leakage) **is below**

$$\left.\frac{\sqrt{g}}{h_y}\tilde{\Gamma}_{ay}\right|_{S_0} = \begin{cases} \left.\left[p_{n_a,a,1}\frac{\sqrt{g}}{h_y}\sqrt{\frac{z_a T_e + T_i}{m_a}} n_a + \min\left\{0, \frac{\sqrt{g}}{h_y} n_a V^{(E)}_{ay}\right\}\right]\right|_{S_0}, & mdf\_fnb = 0 \\ \left.\left[p_{n_a,a,1}\frac{\sqrt{g}}{h_y}\sqrt{\frac{z_a T_e + T_i}{m_a}} n_a + \frac{\sqrt{g}}{h_y} D_{ayx}\frac{1}{h_x}\frac{\partial n_a T_i}{\partial x} + \min\left\{0, \frac{\sqrt{g}}{h_y} n_a V^{(E)}_{ay}\right\}\right]\right|_{S_0}, & mdf\_fnb = 1 \end{cases}$$

$p_{n_a,a,1} = -10^{-3}$

$$S_{n_a,0}^{no\_mdf} = 0$$

$$S_{n_a,1}^{no\_mdf} = p_{n_a,a,1}\frac{\sqrt{g}}{h_y}\sqrt{\frac{z_aT_e+T_i}{m_a}}n_a$$

BCCON=15 is below

$$-\frac{\sqrt{g}}{h_y^2}D_{ay}^n\frac{\partial n_a}{\partial y}\bigg|_{S_0} = \left[c_{n_a,a,2}\frac{\sqrt{g}}{h_y}\sqrt{\frac{z_aT_e+T_i}{m_a}}n_a\right]\bigg|_{S_0}$$

$$c_{n_a,a,2} = -10^{-3}$$

South 1. Core plasma

z_a *=0*

$$\frac{\sqrt{g}}{h_y}\Gamma_{ay}\bigg|_{S_1} = 0$$ *total particle flux (8), dpc*

$$\frac{\sqrt{g}}{h_y}\tilde{\Gamma}_{ay}\bigg|_{S_1} = 0$$

$$z_a \neq 0$$

constant density (1)

$$\frac{\sqrt{g}}{h_y}\tilde{\Gamma}_{ay}\bigg|_{S_1} = \frac{\sqrt{g}}{h_y}(v_{big}n_a\big|_{S_1} - v_{big}n_a)\bigg|_{S_1}, \qquad v_{big} \approx 10^{12},\ \ n_a\big|_{S_1} \text{ is a given density}$$

the total particle flux with constant flux density (8)

$$\frac{\sqrt{g}}{h_y}\tilde{\Gamma}_{ay}\bigg|_{S_1} = p_{n,a,0}\frac{\frac{\sqrt{g}}{h_y}}{\sum_i\left(\frac{\sqrt{g}}{h_y}\right)_{i,-1}\bigg|_{S_1}} + \frac{\sqrt{g}}{h_y}D_{ayx}\frac{1}{h_x}\frac{\partial n_aT_i}{\partial x} + \frac{\sqrt{g}}{h_y}V_{ya}^{(E)}n_a\Bigg|_{S_1}$$

North

z_a *=0*

$$-\frac{\sqrt{g}}{h_y}\Gamma_{ay}\bigg|_N=\left[p_{n_a,a,1}\frac{\sqrt{g}}{h_y}\sqrt{\frac{z_aT_e+T_i}{m_a}}n_a\bigg|_N+\sum_{a=0}^{n_s-1}r_{rec,a,0}c_{nsr}\max\left\{0,\frac{\sqrt{g}}{h_y}\Gamma_{ay}\right\}\right]\Bigg|_N \quad \textit{leakage (10),}\ \textbf{dpc}$$

$p_{n_a,a,1}=0$, $r_{rec,0,0}=0$, $r_{rec,1,0}=1$

$$-\frac{\sqrt{g}}{h_y}\tilde{\Gamma}_{ay}\bigg|_N=\sum_{a=0}^{n_s-1}c_{rec,a,0}c_{nsr}\max\left\{0,\frac{\sqrt{g}}{h_y}\Gamma_{ay}\right\}$$

$\tilde{\Gamma}_{ay}$ was replaced by Γ_{ay}

$c_{rec,0,0}=0$, $c_{rec,1,0}=1$

$z_a\neq 0$

BCCON=9 is below

$$-\frac{\sqrt{g}}{h_y}\tilde{\Gamma}_{ay}\bigg|_N=\begin{cases}\left[\frac{1}{p_{n_a,a,1}}\frac{\sqrt{g}}{h_y}\left(D_{ANa}^n+D_{ANa}^P\left(z_aT_e+T_i\right)\right)n_a\right]\Bigg|_{S_0}, & mdf\_fnb=0\\ \left[\frac{1}{p_{n_a,a,1}}\frac{\sqrt{g}}{h_y}\left(D_{ANa}^n+D_{ANa}^P\left(z_aT_e+T_i\right)\right)n_a-\frac{\sqrt{g}}{h_y}D_{ayx}\frac{1}{h_x}\frac{\partial n_aT_i}{\partial x}-\frac{\sqrt{g}}{h_y}V_{ay}^{(E)}n_a\right]\Bigg|_N, & mdf\_fnb=1\end{cases}$$

$$S_{n_a,0}^{no\_mdf}=0$$
$$S_{n_a,1}^{no\_mdf}=\frac{1}{p_{n_a,a,1}}\frac{\sqrt{g}}{h_y}\left(D_{ANa}^n+D_{ANa}^P\left(z_aT_e+T_i\right)\right)$$

BCCON=10 (leakage) **is below**

$$-\frac{\sqrt{g}}{h_y}\tilde{\Gamma}_{ay}\bigg|_N=\begin{cases}\left[p_{n_a,a,1}\frac{\sqrt{g}}{h_y}\sqrt{\frac{z_aT_e+T_i}{m_a}}n_a-\max\left\{0,\frac{\sqrt{g}}{h_y}n_aV_{ay}^{(E)}\right\}\right]\Bigg|_N, & mdf\_fnb=0\\ \left[p_{n_a,a,1}\frac{\sqrt{g}}{h_y}\sqrt{\frac{z_aT_e+T_i}{m_a}}n_a-\frac{\sqrt{g}}{h_y}D_{ayx}\frac{1}{h_x}\frac{\partial n_aT_i}{\partial x}-\max\left\{0,\frac{\sqrt{g}}{h_y}n_aV_{ay}^{(E)}\right\}\right]\Bigg|_N, & mdf\_fnb=1\end{cases}$$

$$S_{n_a,0}^{no\_mdf}=0$$
$$S_{n_a,1}^{no\_mdf}=p_{n_a,a,1}\frac{\sqrt{g}}{h_y}\sqrt{\frac{z_aT_e+T_i}{m_a}}n_a$$

$$\frac{\sqrt{g}}{h_y^2} D_{ay}^n \frac{\partial n_a}{\partial y}\bigg|_N = \left[c_{n_a,a,2} \frac{\sqrt{g}}{h_y} \sqrt{\frac{z_a T_e + T_i}{m_a}} n_a \right]\bigg|_N$$ **BCCON=15**

$$c_{n_a,a,2} = -10^{-3}$$

North with puffing

z_a *=0*

$$-\frac{\sqrt{g}}{h_y} \tilde{\Gamma}_{ay}\bigg|_N = p_{n,a,0} \frac{\sqrt{g}}{h_y}\bigg|_N + \sum_{a=0}^{n_s-1} \left[c_{rec,a,0} c_{nsr} \max\left\{0, \frac{\sqrt{g}}{h_y} \Gamma_{ay}\right\}\right]\bigg|_N$$ the particle flux per unit area (5)

or $\tilde{\Gamma}_{ay}$ was replaced by Γ_{ay}

$$-\frac{\sqrt{g}}{h_y} \tilde{\Gamma}_{ay}\bigg|_N = \frac{p_{n,a,0}}{\sum_{i-0}^{Nx-1} \left(\frac{\sqrt{g}}{h_y}\right)_{i,N_y}} \frac{\sqrt{g}}{h_y}\Bigg|_N + \sum_{a=0}^{n_s-1} \left[c_{rec,a,0} c_{nsr} \max\left\{0, \frac{\sqrt{g}}{h_y} \Gamma_{ay}\right\}\right]\Bigg|_N$$ ***the total particle flux with constant flux density (8)***

$\tilde{\Gamma}_{ay}$ was replaced by Γ_{ay}

$c_{rec,0,0} = 0$, $c_{rec,1,0} = 1$, $p_{n,0a,0} \sim 10^{21}$,

West

z_a *=0*

$$\frac{\sqrt{g}}{h_x} \Gamma_{ax}\bigg|_W = \left[p_{n,a,1} \frac{\sqrt{g}}{h_x} + \sum_{a=0}^{n_s-1} r_{rec,a,0} c_{nsr} \max\left\{0, -\frac{\sqrt{g}}{h_x} \Gamma_{ax}\right\}\right]\bigg|_W$$ *particle flux per area (5), dpc*

$p_{n_a,a,1} = 0$, $r_{rec,0,0} = 0$, $r_{rec,1,0} = 1$

$$\frac{\sqrt{g}}{h_x} \tilde{\Gamma}_{ax}\bigg|_W = \left[p_{n,a,1} \frac{\sqrt{g}}{h_x} + \sum_{a=0}^{n_s-1} c_{rec,a,0} c_{nsr} \max\left\{0, -\frac{\sqrt{g}}{h_x} \Gamma_{ax}\right\}\right]\bigg|_W$$ It has been replaced $\tilde{\Gamma}_{ax}$ by Γ_{ax}. (5)

$p_{n_a,a,1} = 0$, $c_{rec,0,0} = 0$, $c_{rec,1,0} = 1$

$z_a \neq 0$

$$\frac{\sqrt{g}}{h_x}\Gamma_{ax}\bigg|_W = 10^{20}\frac{\sqrt{g}}{h_x}\frac{\partial n_a}{\partial x}\bigg|_W$$ `sheath density bc (3)` **dpc**

$$\frac{\sqrt{g}}{h_x}\tilde{\Gamma}_{ax}\bigg|_W = \left[-p_{n,a,1}\frac{\sqrt{g}}{h_x}|b_x|\sqrt{\frac{\sum_{b=0}^{n_s-1}p_b}{\sum_{b=0}^{n_s-1}m_b n_b}}n_a - \frac{\sqrt{g}}{h_x}D_{axy}\frac{1}{h_y}\frac{\partial n_a T_i}{\partial y}\right]\Bigg|_W$$ ***sound speed flux bc (14)***

$p_{n,a,1} = +1,$

East

$z_a = 0$

$$-\frac{\sqrt{g}}{h_x}\Gamma_{ax}\bigg|_E = \left[p_{n,a,1}\frac{\sqrt{g}}{h_x} + \sum_{a=0}^{n_s-1}r_{rec,a,0}c_{nsr}\max\left\{0,\frac{\sqrt{g}}{h_x}\Gamma_{ax}\right\}\right]\Bigg|_E$$ `particle flux per area (5), dpc`

$p_{n_a,a,1} = 0$, $r_{rec,0,0} = 0$, $r_{rec,1,0} = 1$

$$-\frac{\sqrt{g}}{h_x}\tilde{\Gamma}_{ax}\bigg|_E = \left[\sum_{a=0}^{n_s-1}c_{rec,a,0}c_{nsr}\max\left\{0,\frac{\sqrt{g}}{h_x}\Gamma_{ax}\right\}\right]\Bigg|_E$$ It has been replaced $\tilde{\Gamma}_{ax}$ by Γ_{ax}. `(5)`

$c_{rec,0,0} = 0$, $c_{rec,1,0} = 1$,

$z_a \neq 0$

$$-\frac{\sqrt{g}}{h_x}\Gamma_{ax}\bigg|_E = -10^{20}\frac{\sqrt{g}}{h_x}\frac{\partial n_a}{\partial x}\bigg|_E$$ `sheath density bc (3)` **dpc**

$$-\frac{\sqrt{g}}{h_x}\tilde{\Gamma}_{ax}\bigg|_E = \left[-p_{n,a,1}\frac{\sqrt{g}}{h_x}|b_x|\sqrt{\frac{\sum_{b=0, z_b\neq 0}^{n_s-1}p_b}{\sum_{b=0, z_b\neq 0}^{n_s-1}m_b n_b}}n_a + \frac{\sqrt{g}}{h_x}D_{axy}\frac{1}{h_y}\frac{\partial n_a T_i}{\partial y}\right]\Bigg|_E$$ ***sound speed flux bc (14)***

$p_{n,a,1} = +1$

ii. Parallel momentum balance for ions

$$S_{m_a} = S_{m_a,0} + S_{m_a,1} u_{\|a} + S_{m_a,2} m_a n_a + S_{m_a,3} m_a n_a u_{\|a}$$

South. Private region

$z_a = 0$,

$$h_z \frac{\sqrt{g}}{h_y} \Gamma_{ay}^m \Big|_{S_0} = \left[p_{m,a,1} p_{m,a,2} h_z \frac{\sqrt{g}}{h_y} - p_{m,a,2} h_z \frac{\sqrt{g}}{h_y} V_{\|a} \right] \Bigg|_{S_0}$$ ***weak velocity (4)*** dpc

$p_{m,a,1} = 0,\ p_{m,a,2} = 10^5$

We had this b.c.

$$h_z \frac{\sqrt{g}}{h_y} \Gamma_{ay}^m \Big|_{S_0} = \left[c_{m,a,3} h_z \frac{\sqrt{g}}{h_y} m_a n_a V_{\|a} + c_{m,a,6} \max\left(0, -h_z \frac{\sqrt{g}}{h_y} \frac{\Gamma_{ay}^{(cor)}}{n_a} \right) m_a n_a V_{\|a} \right] \Bigg|_{S_0}$$, delete term with max !

$$\Gamma_{ay}^{(cor)} = \Gamma_{ay} + \Gamma_{ay}^{(dia)}$$

$c_{m,a,3} = -10^{10}$, we have $c_{m,a,6} = -1$, but not $c_{m,a,6} = 0$

Inserted into b2stbc_phys for bcmom (*12*) as $c_{m,a,3} \to p_{m,a,2} = -10^{10}, c_{m,a,6} \to p_{m,a,1} = -1$

$z_a \neq 0$,

$$h_z \frac{\sqrt{g}}{h_y} \Gamma_{ay}^m \Big|_{S_0} = \left[p_{m,a,1} p_{m,a,2} h_z \frac{\sqrt{g}}{h_y} - p_{m,a,2} h_z \frac{\sqrt{g}}{h_y} V_{\|a} \right] \Bigg|_{S_0}$$ ***weak velocity (4)*** dpc

$p_{m,a,1} = 0,\ p_{m,a,2} = 10^5$

$$h_z \frac{\sqrt{g}}{h_y} \Gamma_{ay}^m \Big|_{S_0} = \left[c_{m,a,3} h_z \frac{\sqrt{g}}{h_y} m_a n_a V_{\|a} + c_{m,a,6} \max\left(0, -h_z \frac{\sqrt{g}}{h_y} \frac{\Gamma_{ay}^{(cor)}}{n_a} \right) m_a n_a V_{\|a} \right] \Bigg|_{S_0}$$, delete term with max !

$$\Gamma_{ay}^{(cor)} = \Gamma_{ay} + \Gamma_{ay}^{(dia)}$$

$c_{m,a,3} = 0$, $c_{m,a,6} = -1$

South 1. Core

$$z_a = 0$$

$$h_z \frac{\sqrt{g}}{h_y} \Gamma_{ay}^m \Big|_{S_1} = \left[p_{m,a,1} p_{m,a,2} h_z \frac{\sqrt{g}}{h_y} - p_{m,a,2} h_z \frac{\sqrt{g}}{h_y} V_{\parallel a} \right]\Bigg|_{S_1}$$ ***weak velocity (4)*** dpc

$p_{m,a,1} = 0,\ p_{m,a,2} = 10^5$

BCMOM=12 $p_{m,a,2} = -10^{10}, p_{m,a,1} = 0$

$$h_z \frac{\sqrt{g}}{h_y} \Gamma_{ay}^m \Big|_{S_1} = P_{m,a,2} h_z \frac{\sqrt{g}}{h_y} m_a n_a V_{\parallel a} \Bigg|_{S_1}$$

$$z_a \neq 0$$

$$h_z \frac{\sqrt{g}}{h_y} \Gamma_{ay}^m \Big|_{S_1} = \left[p_{m,a,1} p_{m,a,2} h_z \frac{\sqrt{g}}{h_y} - p_{m,a,2} \frac{\sqrt{g}}{h_y} V_{\parallel a} \right]\Bigg|_{S_1}$$ ***weak velocity (4)*** dpc

$p_{m,a,1} = 0,\ p_{m,a,2} = 10^5$

BCMOM=14 $p_{m,a,1} = 0$, $p_{m,a,3} = -10^{10}$

$$h_z \frac{\sqrt{g}}{h_y} \Gamma_{ay}^m \Big|_{S_1} = p_{m,a,3} h_z \frac{\sqrt{g}}{h_y} m_a n_a V_{\parallel a} \Bigg|_{S_1} + \left[h_z \frac{\sqrt{g}}{h_y} m_a \langle n_a \rangle (-2) \frac{\langle T_i \rangle B_z}{e z_a} \frac{\partial}{h_x \partial x} \left(\frac{1}{B^2} \right) \langle V_{\parallel a} \rangle c_{drift} + c_{corflux} h_z \frac{\sqrt{g}}{h_y} \right]\Bigg|_{S_1} +$$
$$+ p_{m,a,1} \max\left(0, -h_z \frac{\sqrt{g}}{h_y} \frac{\Gamma_{ay}^{(cor)}}{n_a} \right) m_a n_a V_{\parallel a} \Bigg|_{S_1}$$

The value of the parallel velocity as a function of poloidal coordinate

$$h_z \frac{\sqrt{g}}{h_y} \Gamma_{ay}^m \Big|_{S_1} = v_{big} n_{e0} m_a h_z \frac{\sqrt{g}}{h_y} \left[\frac{\langle B \rangle}{B} p_{m,a,1} - V_{\parallel a} \right]\Bigg|_{S_1},$$ *BCMOM=15*

where $\langle B \rangle = \dfrac{\oint B \sqrt{g} dx}{\oint \sqrt{g} dx}$, the integration is done over inner magnetic surface

$v_{big} \approx 10^{12}$, $n_{e0} = 10^{20}$, $p_{m,a,1}$ is a given value of the average parallel velocity

North

$z_a = 0$,

$$-h_z \frac{\sqrt{g}}{h_y} \Gamma_{ay}^m \Big|_N = \left[p_{m,a,1} p_{m,a,2} h_z \frac{\sqrt{g}}{h_y} - p_{m,a,2} h_z \frac{\sqrt{g}}{h_y} V_{\|a} \right]\Bigg|_N$$

weak velocity (4) dpc

$p_{m,a,1} = 0, \; p_{m,a,2} = 10^5$

$$-h_z \frac{\sqrt{g}}{h_y} \Gamma_{ay}^m \Big|_N = \left[c_{m,a,3} h_z \frac{\sqrt{g}}{h_y} m_a n_a V_{\|a} + c_{m,a,6} \max\left(0, h_z \frac{\sqrt{g}}{h_y} \frac{\Gamma_{ay}^{(cor)}}{n_a} \right) m_a n_a V_{\|a} \right]\Bigg|_N$$

delete term with max !

$c_{m,a,3} = -10^{10}$, we have $c_{m,a,6} = -1$, but not $c_{m,a,6} = 0$

Inserted into b2stbc_phys for BCMOM=12 as $c_{m,a,3} \rightarrow p_{m,a,2} = -10^{10}, c_{m,a,6} \rightarrow p_{m,a,1} = -1$

$z_a \neq 0$,

$$-h_z \frac{\sqrt{g}}{h_y} \Gamma_{ay}^m \Big|_N = \left[p_{m,a,1} p_{m,a,2} h_z \frac{\sqrt{g}}{h_y} - p_{m,a,2} h_z \frac{\sqrt{g}}{h_y} V_{\|a} \right]\Bigg|_N$$

weak velocity (4) dpc

$p_{m,a,1} = 0, \; p_{m,a,2} = 10^5$

$$-h_z \frac{\sqrt{g}}{h_y} \Gamma_{ay}^m \Big|_N = \left[c_{m,a,3} h_z \frac{\sqrt{g}}{h_y} m_a n_a V_{\|a} + c_{m,a,6} \max\left(0, h_z \frac{\sqrt{g}}{h_y} \frac{\Gamma_{ay}^{(cor)}}{n_a} \right) m_a n_a V_{\|a} \right]\Bigg|_N$$

$c_{m,a,3} = 0$, $c_{m,a,6} = -1$

West

$z_a = 0$, $p_{m,a,1} = 0$

$z_a \neq 0$, $p_{m,a,1} = 1$

$$h_z \frac{\sqrt{g}}{h_x} \Gamma_{ax}^m \Big|_W = \left[p_{m,a,1} \frac{u_{big}}{\varepsilon^2} V_{West} m_a n_a - \frac{u_{big}}{\varepsilon^2} h_z \frac{\sqrt{g}}{h_x} |b_x| m_a n_a V_{\|a} \right]\Bigg|_W$$

BCMOM=13

$$V_{West} = -h_z \frac{\sqrt{g}}{h_x} \frac{|b_x|}{b_x} V_{ax}^{(E\times B)} - h_z \frac{\sqrt{g}}{h_x} b_x \sqrt{\frac{\sum_{\substack{b=0\\ z_b\neq 0}}^{n_s-1} p_b}{\sum_{\substack{b=0\\ z_b\neq 0}}^{n_s-1} m_b n_b}}$$

$z_a = 0$ `specified parallel velocity (1) dpc`

$$\frac{\sqrt{g}}{h_x}\Gamma_{ax}^{m}\Big|_W = \left[u_{big} n_{e00} a_{m,a} \frac{\sqrt{g}}{h_x} p_{m,a,1} - u_{big} n_{e00} a_{m,a} \frac{\sqrt{g}}{h_x} V_{\|a}\right]\Bigg|_W, \quad p_{m,a,1} = 0, \quad u_{big} = 10^{18}, \quad n_{e00} = 10^{20} \text{ dpc}$$

$z_a \neq 0$, `sheath velocity (3) dpc`

$p_{m,a,2} < 0.5$, sound_model $= 0$,

$$\frac{\sqrt{g}}{h_x}\Gamma_{ax}^{m}\Big|_W = \left[-\frac{u_{big}}{\varepsilon^2} n_a m_a \frac{\sqrt{g}}{h_x} u_{p_{out}} - \frac{u_{big}}{\varepsilon^2} n_a m_a \frac{\sqrt{g}}{h_x} V_{\|a}\right]\Bigg|_W, \text{ dpc}$$

$p_{m,a,1} = 1, \; p_{m,a,2} = 0, \; u_{big} = 10^{18}, \; n_{e00} = 10^{20}, \; \varepsilon = 10^{-10}$

$$u_{p_{out}} = \max\left\{0, \frac{u_{u_{out}}}{b_x} + \frac{u_a^{(dia)}}{b_x}\right\}, \quad u_{u_{out}} = p_{m,a,1} c_s |b_x|, \quad c_s = \sqrt{\frac{\sum_{\substack{b=0\\ z_b\neq 0}}^{n_s-1} p_b}{\sum_{\substack{b=0\\ z_b\neq 0}}^{n_s-1} m_b n_b}},$$

East

$z_a = 0$, $p_{m,a,1} = 0$

$z_a \neq 0$, $p_{m,a,1} = 1$

$$-h_z \frac{\sqrt{g}}{h_x}\Gamma_{ax}^{m}\Big|_E = \left[p_{m,a,1}\frac{u_{big}}{\varepsilon^2} V_{East} m_a n_a - \frac{u_{big}}{\varepsilon^2} h_z \frac{\sqrt{g}}{h_x} |b_x| m_a n_a V_{\|a}\right]\Bigg|_E \quad \text{BCMOM=13}$$

$$V_{East} = -h_z \frac{\sqrt{g}}{h_x} \frac{|b_x|}{b_x} V_{ax}^{(E\times B)} + h_z \frac{\sqrt{g}}{h_x} b_x \sqrt{\frac{\sum_{\substack{b=0\\ z_b\neq 0}}^{n_s-1} p_b}{\sum_{\substack{b=0\\ z_b\neq 0}}^{n_s-1} m_b n_b}}$$

$z_a = 0$, `specified parallel velocity (1) dpc`

$$-\frac{\sqrt{g}}{h_x}\Gamma_{ax}^{m}\Big|_E = \left[u_{big} n_{e00} a_{m,a} \frac{\sqrt{g}}{h_x} p_{m,a,1} - u_{big} n_{e00} a_{m,a} \frac{\sqrt{g}}{h_x} V_{\parallel a}\right]\Bigg|_E, \quad p_{m,a,1} = 0,\ \ u_{big} = 10^{18},\quad n_{e00} = 10^{20}\ \text{dpc}$$

$z_a \neq 0$, `sheath velocity (3) dpc`

$p_{m,a,2} < 0.5,\ \text{sound\_model} = 0,$

$$-\frac{\sqrt{g}}{h_x}\Gamma_{ax}^{m}\Big|_E = \left[-\frac{u_{big}}{\varepsilon^2} n_a m_a \frac{\sqrt{g}}{h_x} u_{p_{out}} - \frac{u_{big}}{\varepsilon^2} n_a m_a \frac{\sqrt{g}}{h_x} V_{\parallel a}\right]\Bigg|_E,\ \text{dpc}$$

$p_{m,a,1} = 1,\ \ p_{m,a,2} = 0,\ \ u_{big} = 10^{18},\quad n_{e00} = 10^{20},\ \ \varepsilon = 10^{-10}$

$$u_{p_{out}} = \max\left\{0, \frac{u_{u_{out}}}{b_x} - \frac{u_a^{(dia)}}{b_x}\right\},\quad u_{u_{out}} = p_{m,a,1} c_s |b_x|,\quad c_s = \sqrt{\frac{\sum_{\substack{b=0\\ z_b\neq 0}}^{n_s-1} p_b}{\sum_{\substack{b=0\\ z_b\neq 0}}^{n_s-1} m_b n_b}},$$

iii. Current continuity equation

$$S_{ch} = S_{ch,0} + S_{ch,1}\phi + S_{ch,2} n_e + S_{ch,3} n_e \phi$$

South. Private region

$$\frac{\sqrt{g}}{h_y} j_y\Big|_{S_0} = 0\text{, \textbf{dpc}}$$

$$\left.\frac{\sqrt{g}}{h_y} j_y\right|_{S_0} = \begin{cases} 0, & facdrift = 0 \\ \left.\frac{\sqrt{g}}{h_y}\left(\tilde{j}_y^{(dia)} + j_y^{(in)} + \tilde{j}_y^{(vis\|)} + j_y^{(vis\perp)} + \tilde{j}_y^{(visq)} + \tilde{j}_y^{(s)}\right)\right|_{S_0}, & facdrift \neq 0 \end{cases}$$

South 1. Core

$$\left.\frac{\sqrt{g}}{h_y} j_y\right|_{S_1} = \begin{cases} 0, & facdrift = 0 \\ \left.\frac{\sqrt{g}}{h_y}\left(\left\langle \tilde{j}_y^{(dia)} \right\rangle\right)\right|_{S_0}, & facdrift \neq 0 \end{cases} \quad \left\langle \tilde{j}_y^{(dia)} \right\rangle ?\text{, } \mathbf{dpc}$$

$$\left.\frac{\sqrt{g}}{h_y} j_y\right|_{S_1} = \begin{cases} 0, & facdrift = 0 \\ \left.\frac{\sqrt{g}}{h_y}\left(\left\langle \tilde{j}_y^{(dia)} \right\rangle + \left\langle j_y^{(in)} \right\rangle\right)\right|_{S_0}, & facdrift \neq 0 \end{cases}$$

$$\left\langle \tilde{j}_y^{(dia)} \right\rangle = -\frac{B_z \left\langle \sum_{\substack{a=0 \\ z_a \neq 0}}^{n_s-1} n_a \left(z_a T_e + T_i\right) \right\rangle}{h_x} \frac{\partial}{\partial x}\left(\frac{1}{B^2}\right),$$

$$\left\langle j_y^{(in)} \right\rangle = -\frac{B_z}{2} \frac{\partial}{h_x \partial x}\left(\frac{1}{B^2}\right) \sum_{\substack{a=1 \\ z_a \neq 0}}^{n_s-1} m_a n_a \left\langle V_{a\|}^2 \right\rangle,$$

North

$$\left.\frac{\sqrt{g}}{h_y} j_y\right|_{N} = \begin{cases} 0, & facdrift = 0 \\ \left.\frac{\sqrt{g}}{h_y}\left(\left\langle \tilde{j}_y^{(dia)} \right\rangle\right)\right|_{N}, & facdrift \neq 0 \end{cases} \quad \left\langle \tilde{j}_y^{(dia)} \right\rangle ?\text{, } \mathbf{dpc}$$

$$\left.-\frac{\sqrt{g}}{h_y} j_y\right|_{N} = \begin{cases} 0, & facdrift = 0 \\ \left.-\frac{\sqrt{g}}{h_y}\left(\tilde{j}_y^{(dia)} + j_y^{(in)} + \tilde{j}_y^{(vis\|)} + j_y^{(vis\perp)} + \tilde{j}_y^{(visq)} + \tilde{j}_y^{(s)}\right)\right|_{N}, & facdrift \neq 0 \end{cases}$$

West

BCPOT=11 is connected with BCCON=14 when istyle_fchi = 1

$$\left.\frac{\sqrt{g}}{h_x}j_x\right|_W = \frac{\sqrt{g}}{h_x}e\left[-\sum_{a=0}^{n_s-1}z_a\left(p_{n,a,1}|b_x|\sqrt{\frac{\sum\limits_{\substack{b=0\\z_b\neq 0}}^{n_s-1}p_b}{\sum\limits_{\substack{b=0\\z_b\neq 0}}^{n_s-1}m_b n_b}}n_a\right)+(1-\gamma_e)|b_x|n_e\frac{1}{\sqrt{2\pi}}\sqrt{\frac{T_e}{m_e}}\exp\left(-\frac{e\phi}{T_e}\right)\right]\Bigg|_W$$, it is new form

BCPOT=11 when istyle_fchi = 0

$$\left.\frac{\sqrt{g}}{h_x}j_x\right|_W = \frac{\sqrt{g}}{h_x}e\left[\sum_{a=0}^{n_s-1}z_a\Gamma_{ax}+(1-\gamma_e)|b_x|n_e\frac{1}{\sqrt{2\pi}}\sqrt{\frac{T_e}{m_e}}\exp\left(-\frac{e\phi}{T_e}\right)\right]\Bigg|_W$$, it is old form

sheath potential bc (3)

$$\left.\frac{\sqrt{g}}{h_x}j_x\right|_W = \frac{\sqrt{g}}{h_x}e\left[\sum_{a=0}^{n_s-1}z_a\Gamma_{ax}+(1-\gamma_e)|b_x|n_e\frac{1}{\sqrt{2\pi}}\sqrt{\frac{T_e}{m_e}}\exp\left(-\frac{e(\varphi-p_{\varphi,2})}{T_e}\right)\right]\Bigg|_W$$, dpc

East

BCPOT=11 is connected with BCCON=14 when istyle_fchi = 1

$$-\left.\frac{\sqrt{g}}{h_x}j_x\right|_E = \frac{\sqrt{g}}{h_x}e\left[-\sum_{a=0}^{n_s-1}z_a\left(p_{n,a,1}|b_x|\sqrt{\frac{\sum\limits_{\substack{b=0\\z_b\neq 0}}^{n_s-1}p_b}{\sum\limits_{\substack{b=0\\z_b\neq 0}}^{n_s-1}m_b n_b}}n_a\right)+(1-\gamma_e)|b_x|n_e\frac{1}{\sqrt{2\pi}}\sqrt{\frac{T_e}{m_e}}\exp\left(-\frac{e\phi}{T_e}\right)\right]\Bigg|_E$$, it is new form

BCPOT=11 when istyle_fchi = 0

$$-\left.\frac{\sqrt{g}}{h_x}j_x\right|_E = \frac{\sqrt{g}}{h_x}e\left[-\sum_{a=0}^{n_s-1}z_a\Gamma_{ax}+(1-\gamma_e)|b_x|n_e\frac{1}{\sqrt{2\pi}}\sqrt{\frac{T_e}{m_e}}\exp\left(-\frac{e\phi}{T_e}\right)\right]\Bigg|_E$$, it is old form

$c_{ch,7}=1-\gamma_e$, $c_{ch,6}=1$

sheath potential bc (3)

$$-\left.\frac{\sqrt{g}}{h_x}j_x\right|_E = \frac{\sqrt{g}}{h_x}e\left[-\sum_{a=0}^{n_s-1}z_a\Gamma_{ax}+(1-\gamma_e)|b_x|n_e\frac{1}{\sqrt{2\pi}}\sqrt{\frac{T_e}{m_e}}\exp\left(-\frac{e(\varphi-p_{\varphi,2})}{T_e}\right)\right]\Bigg|_E$$, dpc

iv. Energy balance equation for electrons

$$S_{h_e}=S_{h_e,0}+S_{h_e,1}T_e+S_{h_e,2}n_e+S_{h_e,3}n_eT_e$$

South. Private region

$$\frac{\sqrt{g}}{h_y}\tilde{q}_{ey}\Big|_{S_0} = \left[\left(c_{h_e,2}\frac{\sqrt{g}}{h_y}\sqrt{\frac{T_e}{m_e}}\right)T_e n_e + c_{h_e,7}\max\left(0,-\frac{\sqrt{g}}{h_y}\tilde{f}_{ey}\right)T_e - q_{ex}^{P.Sch}\right]\Big|_{S_0},\qquad \text{bcene=11}$$

$c_{h_e,2} = -10^{-4}$, $c_{h_e,7} = -1$, $c_{h_e,2} \to p_{h_e,2} = -10^{-4}$, $c_{h_e,7} \to p_{h_e,1} = -1$

$$-\frac{\sqrt{g}}{h_y}k_{ey}\frac{\partial T_e}{h_y \partial y}\Big|_{S_0} = \left[\left(p_{h_e,1}\frac{\sqrt{g}}{h_y}\sqrt{\frac{T_e}{m_e}}\right)T_e n_e\right]\Big|_{S_0},\qquad \text{bcene=22}$$

BCENE=9 (electron temperature decay length)

$$\frac{\sqrt{g}}{h_y}\tilde{q}_{ey}\Big|_{S_0} = \begin{cases} \left[\frac{5}{2}\sum_{a=0}^{n_s-1} z_a S_{n_a,0}^{no\_mdf} T_e - \frac{\sqrt{g}}{h_y}\frac{\kappa_e^{(AN)}}{p_{h_e,1}}T_e + \frac{5}{2}\sum_{a=0}^{n_s-1} z_a S_{n_a,1}^{no\_mdf} n_a T_e\right]\Big|_{S_0}, & mdf\_fhe = 0 \\ \left[\frac{5}{2}\sum_{a=0}^{n_s-1} z_a S_{n_a,1}^{no\_mdf} T_e - \frac{\sqrt{g}}{h_y}\frac{\kappa_e^{(AN)}}{p_{h_e,1}}T_e + \frac{5}{2}\sum_{a=0}^{n_s-1} z_a S_{n_a,1}^{no\_mdf} n_a T_e - q_{ey}^{P.Sch} + \frac{\sqrt{g}}{h_y}\frac{3}{2}V_{ey}^{(E)} n_e T_e\right]\Big|_{S_0}, & mdf\_fhe = 1 \end{cases}$$

South 1. Core

$$\frac{\sqrt{g}}{h_y}\tilde{q}_{ey}\Big|_{S_1} = \left[\left(c_{h_e,0}\frac{\sqrt{g}}{h_y} + c_{h_e,1}\frac{\sqrt{g}}{h_y}T_e\right)\right]\Big|_{S_1}$$

$c_{h_e,0} = 4\cdot 10^{13}$, $c_{h_e,a,1} = -10^{30}$,

`total electron energy flux (8)`

$$\frac{\sqrt{g}}{h_y}q_{ey}\Big|_{S_1} = p_{h_e,1}\frac{\frac{\sqrt{g}}{h_y}}{\sum_{core}\left(\frac{\sqrt{g}}{h_y}\right)}\Bigg|_{S_1},\quad p_{h_e,1} = 8\cdot 10^5\quad \text{, dpc}$$

North

$$-\frac{\sqrt{g}}{h_y}\tilde{q}_{ey}\Big|_{N} = \left[\left(c_{h_e,2}\frac{\sqrt{g}}{h_y}\sqrt{\frac{T_e}{m_e}}\right)T_e n_e + c_{h_e,7}\max\left(0,\frac{\sqrt{g}}{h_y}\tilde{f}_{ey}\right)T_e + q_{ey}^{P.Sch}\right]\Big|_{N},\qquad \text{bcene=11}$$

$$c_{h_e,2} = -10^{-4},\ c_{h_e,7} = -1$$

$$\frac{\sqrt{g}}{h_y} k_{ey} \frac{\partial T_e}{h_y \partial y}\bigg|_N = \left[\left(p_{h_e,1} \frac{\sqrt{g}}{h_y} \sqrt{\frac{T_e}{m_e}}\right) T_e n_e \right]\bigg|_N ,$$ bcene=22

BCENE=9 (electron temperature decay length) $p_{h_e,1} = 10^{-2}$

$$-\frac{\sqrt{g}}{h_y} \tilde{q}_{ey}\Big|_N = \begin{cases} \left[\frac{5}{2}\sum_{a=0}^{n_s-1} z_a S_{n_a,0}^{no\_mdf} T_e - \frac{\sqrt{g}}{h_y} \frac{\kappa_e^{(AN)}}{p_{h_e,1}} T_e + \frac{5}{2}\sum_{a=0}^{n_s-1} z_a S_{n_a,1}^{no\_mdf} n_a T_e \right]\Bigg|_N , & mdf\_fhe = 0 \\ \left[\frac{5}{2}\sum_{a=0}^{n_s-1} z_a S_{n_a,0}^{no\_mdf} T_e - \frac{\sqrt{g}}{h_y} \frac{\kappa_e^{(AN)}}{p_{h_e,1}} T_e + \frac{5}{2}\sum_{a=0}^{n_s-1} z_a S_{n_a,1}^{no\_mdf} n_a T_e + q_{ey}^{P.Sch} - \frac{\sqrt{g}}{h_y}\frac{3}{2} V_{ey}^{(E)} n_e T_e \right]\Bigg|_N , & mdf\_fhe = 1 \end{cases}$$

West

bcene_15_style.eq.0 is connected with BCCON=14

$$\frac{\sqrt{g}}{h_x} \tilde{q}_{ex}\Big|_W = \left[-\left(\frac{1+\gamma}{1-\gamma} T_e + e\varphi\right) \max\left(0, \frac{\sqrt{g}}{h_x} \sum_{a=0}^{n_s-1} z_a p_{n,a,1} |b_x| \sqrt{\frac{\sum_{b=0, z_b\neq 0}^{n_s-1} p_b}{\sum_{b=0, z_b\neq 0}^{n_s-1} m_b n_b}} n_a + \frac{\sqrt{g}}{h_x} \frac{\tilde{j}_\parallel}{e}\right) - \frac{\sqrt{g}}{h_x} q_{ex}^{P.Sch} \right]\Bigg|_W$$ bcene=15

$$\frac{\sqrt{g}}{h_x} \tilde{q}_{ex}\Big|_W = \left[-\left(\frac{1+\gamma}{1-\gamma} + \frac{e\varphi}{T_e}\right) \max\left(0, \frac{\sqrt{g}}{h_x} \sum_{a=0}^{n_s-1} z_a p_{n,a,1} |b_x| \sqrt{\frac{\sum_{b=0, z_b\neq 0}^{n_s-1} p_b}{\sum_{b=0, z_b\neq 0}^{n_s-1} m_b n_b}} n_a + \frac{\sqrt{g}}{h_x} \frac{\tilde{j}_\parallel}{e}\right) T_e - \frac{\sqrt{g}}{h_x} q_{ex}^{P.Sch} \right]\Bigg|_W$$ bcene=15

bcene_15_style.eq.1

$$\frac{\sqrt{g}}{h_x} \tilde{q}_{ex}\Big|_W = \left[-\frac{\sqrt{g}}{h_x} |b_x| \left((1+\gamma) T_e + (1-\gamma) e\varphi\right) n_e \frac{1}{\sqrt{2\pi}} \sqrt{\frac{T_e}{m_e}} \exp\left(-\frac{e\phi}{T_e}\right) - \frac{\sqrt{g}}{h_x} q_{ex}^{P.Sch} \right]\Bigg|_W$$ bcene=15

sheath electron (3)

$p_{m,a_{main},2} < 0.5,\ \text{sound\_model} = 0,$

$if \quad p_{h_e,1} > 0 \quad then$

$$\frac{\sqrt{g}}{h_x} q_{ex}\Big|_W = \left[p_{h_e,1} + e\frac{\varphi - p_{\varphi,2}}{T_e} \right] \frac{\sqrt{g}}{h_x} \nu_{e_{out}} T_e - \frac{\sqrt{g}}{h_x} q_{ex}^{P.Sch} \Bigg|_W , \quad p_{h_e,1} = 0.9, \quad p_{\varphi,2} = 0,$$

$$\nu_{e_{out}} = \min\left\{ 0, -u_{u_{out}} n_e - \frac{j_x}{e} \right\}, \quad u_{u_{out}} = p_{m,a_{main},1} c_s |b_x|, \quad c_s = \sqrt{\frac{\sum\limits_{\substack{b=0 \\ z_b \neq 0}}^{n_s-1} p_b}{\sum\limits_{\substack{b=0 \\ z_b \neq 0}}^{n_s-1} m_b n_b}},$$

endif

East

bcene_15_style.eq.0 is connected with BCCON=14

$$-\frac{\sqrt{g}}{h_x}\tilde{q}_{ex}\Big|_E = \left[-\left(\frac{1+\gamma}{1-\gamma} T_e + e\varphi \right) \max\left(0, \frac{\sqrt{g}}{h_x} \sum_{a=0}^{n_s-1} z_a p_{n,a,1} |b_x| \sqrt{\frac{\sum\limits_{\substack{b=0 \\ z_b \neq 0}}^{n_s-1} p_b}{\sum\limits_{\substack{b=0 \\ z_b \neq 0}}^{n_s-1} m_b n_b}} n_a - \frac{\sqrt{g}}{h_x} \frac{\tilde{j}_\parallel}{e} \right) + \frac{\sqrt{g}}{h_x} q_{ex}^{P.Sch} \right]\Bigg|_E \quad \text{bcene=15}$$

$$-\frac{\sqrt{g}}{h_x}\tilde{q}_{ex}\Big|_E = \left[-\left(\frac{1+\gamma}{1-\gamma} + \frac{e\varphi}{T_e} \right) \max\left(0, \frac{\sqrt{g}}{h_x} \sum_{a=0}^{n_s-1} z_a p_{n,a,1} |b_x| \sqrt{\frac{\sum\limits_{\substack{b=0 \\ z_b \neq 0}}^{n_s-1} p_b}{\sum\limits_{\substack{b=0 \\ z_b \neq 0}}^{n_s-1} m_b n_b}} n_a - \frac{\sqrt{g}}{h_x} \frac{\tilde{j}_\parallel}{e} \right) T_e + \frac{\sqrt{g}}{h_x} q_{ex}^{P.Sch} \right]\Bigg|_E \quad \text{bcene=15}$$

bcene_15_style.eq.1

$$-\frac{\sqrt{g}}{h_x}\tilde{q}_{ex}\Big|_E = \left[-\frac{\sqrt{g}}{h_x} |b_x| \left((1+\gamma) T_e + (1-\gamma) e\varphi \right) n_e \frac{1}{\sqrt{2\pi}} \sqrt{\frac{T_e}{m_e}} \exp\left(-\frac{e\phi}{T_e} \right) + \frac{\sqrt{g}}{h_x} q_{ex}^{P.Sch} \right]\Bigg|_E \quad \text{bcene=15}$$

`sheath electron (3)`

$p_{m,a_{main},2} < 0.5,\ \text{sound\_model} = 0,$

$if \quad p_{h_e,1} > 0 \quad then$

$$-\frac{\sqrt{g}}{h_x}\tilde{q}_{ex}\Big|_E = -\left[p_{h_e,1} + e\frac{\varphi - p_{\varphi,2}}{T_e}\right]\frac{\sqrt{g}}{h_x}\nu_{e_{out}}T_e + \frac{\sqrt{g}}{h_x}q_{ex}^{P.Sch}\Bigg|_E, \quad p_{h_e,1} = 0.9, \quad p_{\varphi,2} = 0,$$

$$\nu_{e_{out}} = \max\left\{0, u_{u_{out}}n_e - \frac{j_x}{e}\right\}, \quad u_{u_{out}} = p_{m,a_{main},1}c_s|b_x|, \quad c_s = \sqrt{\frac{\sum\limits_{\substack{b=0\\z_b\neq0}}^{n_s-1}p_b}{\sum\limits_{\substack{b=0\\z_b\neq0}}^{n_s-1}m_b n_b}},$$

endif

v. Energy balance equation for ions

$$S_{h_i} = S_{h_i,0} + S_{h_i,1}T_i + S_{h_i,2}n_i + S_{h_i,3}n_iT_i$$

South. Private region

$p_{h_i,1} = 10^{-2}$, $e_{rec,0} = 0$, $e_{rec,1} = 0.3$, $r_{rec,0,0} = 0$, $r_{rec,1,0} = 1$

BCENE=9 (electron temperature decay length)

$$\frac{\sqrt{g}}{h_y}\tilde{q}_{iy}\Big|_{S_0} = \begin{cases} \left[\begin{array}{l} \frac{5}{2}\sum\limits_{a=0}^{n_s-1}S_{n_a,0}^{no\_mdf}T_i - \frac{\sqrt{g}}{h_y}\frac{\kappa_i^{(AN)}}{p_{h_i,1}}T_i + \frac{5}{2}\sum\limits_{a=0}^{n_s-1}S_{n_a,1}^{no\_mdf}n_aT_i + \\ +\sum\limits_{a=0}^{n_s-1}\max\left\{e_{rec,a}\left(\frac{3}{2}+B_{oRiS}\right)(z_{a,f}T_{e,f}+T_{i,f}),\left(\frac{3}{2}+B_{oRiS}\right)T_{Surf}k_B\right\}r_{rec,a,0}c_{nsr}\max\left\{0,-\frac{\sqrt{g}}{h_y}\Gamma_{ay}\right\} \end{array}\right]\Bigg|_{S_0}, & mdf\_fhi = 0 \\ \left[\begin{array}{l} \frac{5}{2}\sum\limits_{a=0}^{n_s-1}S_{n_a,0}^{no\_mdf}T_i - \frac{\sqrt{g}}{h_y}\frac{\kappa_i^{(AN)}}{p_{h_i,1}}T_i + \frac{5}{2}\sum\limits_{a=0}^{n_s-1}S_{n_a,1}^{no\_mdf}n_aT_i - q_{iy}^{P.Sch} + \frac{\sqrt{g}}{h_y}\frac{3}{2}\sum\limits_{a=0}^{n_s-1}V_{ay}^{(E)}n_aT + \\ +\sum\limits_{a=0}^{n_s-1}\max\left\{e_{rec,a}\left(\frac{3}{2}+B_{oRiS}\right)(z_{a,f}T_{e,f}+T_{i,f}),\left(\frac{3}{2}+B_{oRiS}\right)T_{Surf}k_B\right\}r_{rec,a,0}c_{nsr}\max\left\{0,-\frac{\sqrt{g}}{h_y}\Gamma_{ay}\right\} \end{array}\right]\Bigg|_{S_0}, & mdf\_fhi = 1 \end{cases}$$

$$\frac{\sqrt{g}}{h_y}\tilde{q}_{iy}\Big|_{S_0} = p_{h_i,2}\sum_{\substack{a=0\\z_a\neq0}}^{n_s-1}\left(\frac{\sqrt{g}}{h_y}n_a\sqrt{\frac{z_aT_e+T_i}{m_a}}\right)T_i\Bigg|_{S_0} + p_{h_i,1}\max\left(0,-\frac{\sqrt{g}}{h_y}\tilde{f}_{iy}\right)T_i\Bigg|_{S_0}, \qquad \text{bceni=21}$$

$p_{h_i,1} = -1$, $p_{h_i,2} = -10^{-2}$

$$-\frac{\sqrt{g}}{h_y}k_{iy}\frac{\partial T_i}{h_y\partial y}\bigg|_{S_0} = p_{h_i,1}\sum_{\substack{a=0\\ z_a\neq 0}}^{n_s-1}\left(\frac{\sqrt{g}}{h_y}n_a\sqrt{\frac{z_aT_e+T_i}{m_a}}\right)T_i\Bigg|_{S_0},$$ bceni=22

$$\frac{\sqrt{g}}{h_y}\tilde{q}_{iy}\Big|_{S_0} = \sum_{\substack{a=0\\ z_a\neq 0}}^{n_s-1}\left(c_{h_i,a,2}\frac{\sqrt{g}}{h_y}\frac{n_a}{n_i}\sqrt{\frac{z_aT_e+T_i}{m_a}}\right)n_iT_i\Bigg|_{S_0} + c_{h_i,0,7}\max\left(0,-\frac{\sqrt{g}}{h_y}\tilde{f}_{iy}\right)T_i\Bigg|_{S_0},$$ b2stbc_spb

$c_{h_i,a,2} = -10^{-2}$

South 1. Core

total ion energy flux (8)

$$\frac{\sqrt{g}}{h_y}\tilde{q}_{iy}\Big|_{S_1} = \left[p_{h_i,1}\frac{\dfrac{\sqrt{g}}{h_y}}{\displaystyle\sum_{core}\left(\frac{\sqrt{g}}{h_y}\right)} - q_{iy}^{P.Sch} + \frac{3}{2}\frac{\left(j_y^{(AN)} + j_y^{(in)} + \tilde{j}_y^{(vis\|)} + \tilde{j}_y^{(visq)} + j_y^{(vis\perp)}\right)}{e}T_i\right]\Bigg|_{S_1},$$

$$\frac{\sqrt{g}}{h_y}\tilde{q}_{iy}\Big|_{S_1} = \sum_{a=0}^{n_s-1}\left[\left(c_{h_i,a,0}\frac{\sqrt{g}}{h_y}\frac{n_a}{n_i} + c_{h_i,a,1}\frac{\sqrt{g}}{h_y}\frac{n_a}{n_i}T_i\right)\right]\Bigg|_{S_1}$$

$c_{h_i,a,0} = 4\cdot 10^{13}, c_{h_i,a,1} = -10^{30}, a = 0,1,$

North

BCENE=9 (electron temperature decay length)

$p_{h_i,1} = 10^{-2}$, $e_{rec,0} = 0$, $e_{rec,1} = 0.3$, $r_{rec,0,0} = 0$, $r_{rec,1,0} = 1$

$$-\frac{\sqrt{g}}{h_y}\tilde{q}_{iy}\Big|_N=\begin{cases}\left[\begin{aligned}&\frac{5}{2}\sum_{a=0}^{n_s-1}S_{n_a,0}^{no\_mdf}T_i-\frac{\sqrt{g}}{h_y}\frac{\kappa_i^{(AN)}}{p_{h_i,1}}T_i+\frac{5}{2}\sum_{a=0}^{n_s-1}S_{n_a,1}^{no\_mdf}n_aT_i+\\&+\sum_{a=0}^{n_s-1}\max\left\{e_{rec,a}\left(\frac{3}{2}+B_{oRiS}\right)\left(z_{a,f}T_{e,f}+T_{i,f}\right),\left(\frac{3}{2}+B_{oRiS}\right)T_{Surf}k_B\right\}r_{rec,a,0}c_{nsr}\max\left\{0,-\frac{\sqrt{g}}{h_y}\Gamma_{ay}\right\}\end{aligned}\right]\Bigg|_N, & mdf\_fhi=0\\ \left[\begin{aligned}&\frac{5}{2}\sum_{a=0}^{n_s-1}S_{n_a,0}^{no\_mdf}T_i-\frac{\sqrt{g}}{h_y}\frac{\kappa_i^{(AN)}}{p_{h_i,1}}T_i+\frac{5}{2}\sum_{a=0}^{n_s-1}S_{n_a,1}^{no\_mdf}n_aT_i+q_{iy}^{P.Sch}-\frac{\sqrt{g}}{h_y}\frac{3}{2}\sum_{a=0}^{n_s-1}V_{ay}^{(E)}n_aT+\\&+\sum_{a=0}^{n_s-1}\max\left\{e_{rec,a}\left(\frac{3}{2}+B_{oRiS}\right)\left(z_{a,f}T_{e,f}+T_{i,f}\right),\left(\frac{3}{2}+B_{oRiS}\right)T_{Surf}k_B\right\}r_{rec,a,0}c_{nsr}\max\left\{0,-\frac{\sqrt{g}}{h_y}\Gamma_{ay}\right\}\end{aligned}\right]\Bigg|_N, & mdf\_fhi=1\end{cases}$$

$$-\frac{\sqrt{g}}{h_y}\tilde{q}_{iy}\Big|_N=p_{h_i,2}\sum_{\substack{a=0\\z_a\neq 0}}^{n_s-1}\left(\frac{\sqrt{g}}{h_y}n_a\sqrt{\frac{z_aT_e+T_i}{m_a}}\right)T_i\Bigg|_N+p_{h_i,1}\max\left(0,-\frac{\sqrt{g}}{h_y}\tilde{f}_{iy}\right)T_i\Bigg|_N,$$

bceni=21

$p_{h_i,1}=-1$, $p_{h_i,2}=-10^{-2}$

$$\frac{\sqrt{g}}{h_y}k_{iy}\frac{\partial T_i}{h_y\partial y}\bigg|_N=p_{h_i,1}\sum_{\substack{a=0\\z_a\neq 0}}^{n_s-1}\left(\frac{\sqrt{g}}{h_y}n_a\sqrt{\frac{z_aT_e+T_i}{m_a}}\right)T_i\Bigg|_N, \qquad \text{bceni=22}$$

$$-\frac{\sqrt{g}}{h_y}\tilde{q}_{iy}\Big|_N=\sum_{\substack{a=0\\z_a\neq 0}}^{n_s-1}\left(c_{h_i,a,2}\frac{\sqrt{g}}{h_y}\frac{n_a}{n_i}\sqrt{\frac{z_aT_e+T_i}{m_a}}\right)n_iT_i\Bigg|_N+c_{h_i,0,7}\max\left(0,-\frac{\sqrt{g}}{h_y}\tilde{f}_{iy}\right)T_i\Bigg|_N, \qquad \text{b2stbc\_spb}$$

$c_{h_i,a,2}$ = -10^{-2}

West

$$\frac{\sqrt{g}}{h_x}\tilde{q}_{ix}\Big|_W=-p_{h_{i,1}}\frac{\sqrt{g}}{h_x}|b_x|\sqrt{\frac{\sum\limits_{\substack{b=0\\z_b\neq 0}}^{n_s-1}p_b}{\sum\limits_{\substack{b=0\\z_b\neq 0}}^{n_s-1}m_bn_b}}T_i\sum_{\substack{a=0\\z_a\neq 0}}^{n_s-1}p_{n,a,1}n_a-\frac{\sqrt{g}}{h_x}q_{ix}^{P.Sch}\Bigg|_W \qquad \text{bceni=15}$$

$p_{h_i,1} = +1.5$

`sheath ion (3)` BCENI=3

$p_{m,a_{main},2} = 0 < 0.5,\ \text{sound\_model} = 0,$

$$\frac{\sqrt{g}}{h_x}\tilde{q}_{ix}\Big|_W = \left[\begin{array}{l} \sum\limits_{z_a \neq 0}\left(\frac{\sqrt{g}}{h_x}\frac{p_{h_i,2}}{2}u_{u_{out}}m_a n_a\left(u_{p_{out}}^2 + \left(u_{a,x}^{(dia)}\right)^2\right)_i - \frac{\sqrt{g}}{h_x}p_{h_i,1}u_{u_{out}}n_a T_i\right) + \\ + \sum\limits_{a=0}^{n_s-1}\max\left\{c_{rec,a,1}\left(\frac{3}{2}+B_{oRiS}\right)\left(z_{a,f}T_{e,f}+T_{i,f}\right),\left(\frac{3}{2}+B_{oRiS}\right)T_{Surf}k_B\right\}r_{rec,a,0}c_{nsr}\max\left\{0,-\frac{\sqrt{g}}{h_x}\Gamma_{ax}\right\} \end{array}\right]\Bigg|_W$$

$$u_{a,x}^{(dia)} = V_{ax}^{(E)} + \tilde{V}_{xa}^{(dia)}$$

$$u_{p_{out}} = \max\left\{0, \frac{u_{u_{out}}}{|b_x|} + \frac{|B_z|}{|B_x|}u_{1,x}^{(dia)}\right\}$$

$$u_{u_{out}} = p_{m,a_{main},1}c_s|b_x|, \qquad c_s = \sqrt{\frac{\sum\limits_{\substack{b=0\\ z_b\neq 0}}^{n_s-1}p_b}{\sum\limits_{\substack{b=0\\ z_b\neq 0}}^{n_s-1}m_b n_b}}, \qquad p_{m,a_{main},1} = 1.0$$

if neutrals_namelist.eq.1 .or. use_eirene.ne.0 then $e_{rec,0} = 0$, $e_{rec,1} = 0.3$, $r_{rec,0} = 0$, $r_{rec,1} = 1$ **else** $c_{rec,0,0} = 0$, $c_{rec,1,0} = 1$, $c_{rec,0,1} = 0$, $c_{rec,1,1} = 0.3$

$p_{h_i,1} = 2.5$ **but it should be** $p_{h_i,1} = 1.5$

$p_{h_i,2} = 0$ **without drifts**

$p_{h_i,2} = 10^{-3}$ **with drifts but it should be** $p_{h_i,2} = 0$

East

$$-\frac{\sqrt{g}}{h_x}\tilde{q}_{ix}\Big|_E = -p_{h_{i},1}\frac{\sqrt{g}}{h_x}|b_x|\sqrt{\frac{\sum\limits_{\substack{b=0\\ z_b\neq 0}}^{n_s-1}p_b}{\sum\limits_{\substack{b=0\\ z_b\neq 0}}^{n_s-1}m_b n_b}}T_i\sum_{\substack{a=0\\ z_a\neq 0}}^{n_s-1}p_{n,a,1}n_a + \frac{\sqrt{g}}{h_x}q_{ix}^{P.Sch}\Bigg|_E \qquad \text{BCENI=15}$$

$p_{h_i,1} = +1.5$

`sheath ion (3)` BCENI=3

$p_{m,a_{main},2} = 0 < 0.5,\ \text{sound\_model} = 0,$

$$-\frac{\sqrt{g}}{h_x}\tilde{q}_{ix}\Big|_E = \left[\begin{array}{l}\sum_{z_a\neq 0}\left(-\frac{\sqrt{g}}{h_x}\frac{p_{h_i,2}}{2}u_{u_{out}}m_a n_a\left(u_{p_{out}}^2+\left(u_{a,x}^{(dia)}\right)^2\right)-\frac{\sqrt{g}}{h_x}p_{h_i,1}u_{u_{out}}n_a T_i\right)+\\ +\sum_{a=0}^{n_s-1}\max\left\{c_{rec,a,1}\left(\frac{3}{2}+B_{oRiS}\right)\left(z_{a,f}T_{e,f}+T_{i,f}\right),\left(\frac{3}{2}+B_{oRiS}\right)T_{Surf}k_B\right\}c_{rec,a,0}c_{nsr}\max\left\{0,\frac{\sqrt{g}}{h_x}\Gamma_{ax}\right\}\end{array}\right]\Bigg|_E$$

$$u_{a,x}^{(dia)} = V_{ax}^{(E)} + \tilde{V}_{xa}^{(dia)}$$

$$u_{p_{out}} = \max\left\{0,\frac{u_{u_{out}}}{|b_x|}-\frac{|B_z|}{|B_x|}u_{1,x}^{(dia)}\right\}$$

$$u_{u_{out}} = p_{m,a_{main},1}c_s|b_x|,\quad c_s = \sqrt{\frac{\sum\limits_{\substack{b=0\\ z_b\neq 0}}^{n_s-1}p_b}{\sum\limits_{\substack{b=0\\ z_b\neq 0}}^{n_s-1}m_b n_b}},\quad p_{m,a_{main},1} = 1.0$$

if neutrals_namelist.eq.1 .or. use_eirene.ne.0 then $e_{rec,0}=0$, $e_{rec,1}=0.3$, $r_{rec,0}=0$, $r_{rec,1}=1$ **else** $c_{rec,0,0}=0$, $c_{rec,1,0}=1$, $c_{rec,0,1}=0$, $c_{rec,1,1}=0.3$

$p_{h_i,1}=2.5$ **but it should be** $p_{h_i,1}=1.5$

$p_{h_i,2}=0$ **without drifts**

$p_{h_i,2}=10^{-3}$ **with drifts but it should be** $p_{h_i,2}=0$

Appendix

Old form of $\left(\nabla\pi_a^{\|}\right)_{\|}$ was

$$\left(\nabla\pi_a^{\|}\right)_{\|}=\begin{cases}0, & a\neq 1\\ -\dfrac{4}{3}b_xB^{\frac{3}{2}}\dfrac{\partial}{h_x\partial x}\left(\dfrac{c_{\eta_{ax}^{(CL)}}c_{\eta_{ax}^{(CL)}}^{(Luciani)}\eta_{CLax}b_x}{B^2}\dfrac{\partial\left(B^{\frac{1}{2}}\left(V_{a\|}+0\cdot\dfrac{B}{B_x}V_{ax}^{(dia)}+0\cdot\dfrac{B}{B_x}V_x^{(E)}\right)\right)}{h_x\partial x}\right)- & a=1\\ -0\cdot B^{\frac{3}{2}}b_x\dfrac{\partial}{h_x\partial x}\left(\dfrac{b_x}{\nu_{aa}B^2}\dfrac{\partial\left(B^{\frac{1}{2}}\left(q_{a\|}^{(0)}+\dfrac{B}{B_x}q_{ax}^{(dia)}\right)\right)}{h_x\partial x}\right) & \end{cases}$$

Contribution from anomalous viscosity was contained in l.h.s.

Because of $\eta_{ANa}<<\tilde{\tilde{\eta}}_{CLax}$,$\eta_{ax}\approx\tilde{\tilde{\eta}}_{CLax}$ and taking into account $B_xh_xh_y=const$

$$-\frac{4}{3}b_xB^{\frac{3}{2}}\frac{\partial}{h_x\partial x}\left(\frac{c_{\eta_{ax}^{(CL)}}c_{\eta_{ax}^{(CL)}}^{(Luciani)}\eta_{CLax}b_x}{B^2}\frac{\partial\left(B^{\frac{1}{2}}V_{a\|}\right)}{h_x\partial x}\right)=\frac{\partial}{h_z\sqrt{g}\partial x}\left(\frac{h_z\sqrt{g}}{h_x}\left(\frac{4}{3}\eta_{ax}\frac{\partial\ln h_z}{h_x\partial x}V_{\|a}-\frac{4}{3}\eta_{ax}\frac{\partial V_{\|a}}{h_x\partial x}\right)\right)- \quad ,\ a=1$$

$$-\frac{\partial}{h_z\sqrt{g}\partial x}\left(\frac{h_z\sqrt{g}}{h_x}\frac{4}{3}\eta_{ax}\frac{\partial\ln\left(h_zB^{\frac{1}{2}}\right)}{B^{\frac{1}{2}}h_x\partial x}\right)B^{\frac{1}{2}}V_{\|a}$$

,

New form of $\left(\nabla\pi_a^{\|}\right)_{\|}$

$$\left(\nabla\pi_a^{\|}\right)_{\|}=\begin{cases}0, & a\neq 1,\\ \dfrac{\partial}{h_z\sqrt{g}\partial x}\left(\dfrac{h_z\sqrt{g}}{h_x}\left(\dfrac{4}{3}\eta_{ax}\dfrac{\partial\ln h_z}{h_x\partial x}V_{\|a}-\dfrac{4}{3}\eta_{ax}\dfrac{\partial V_{\|a}}{h_x\partial x}\right)\right)-\dfrac{\partial}{h_z\sqrt{g}\partial x}\left(\dfrac{h_z\sqrt{g}}{h_x}\dfrac{4}{3}\eta_{ax}\dfrac{\partial\ln\left(h_zB^{\frac{1}{2}}\right)}{B^{\frac{1}{2}}h_x\partial x}\right)B^{\frac{1}{2}}V_{\|a} & ,\ a=1\end{cases}$$

About scaling of residuals

$$\frac{\sum_{i=0}^{n_x-1}\sum_{j=0}^{n_y-1}\left|r_{n_{a,i,j}}\right|}{\sum_{i=0}^{n_x-1}\sum_{j=0}^{n_{xy}-1}\sqrt{g_{i,j}}\,n_{a,i,j}},$$

$$\frac{\sum_{i=0}^{n_x-1}\sum_{j=0}^{n_y-1}\left|r_{u_{a,i,j}}\right|}{\sum_{i=0}^{n_x-1}\sum_{j=0}^{n_{xy}-1}\sqrt{g_{i,j}}\,n_{a,i,j}\sqrt{m_aT_i}}$$

$$\frac{\sum_{i=0}^{n_x-1}\sum_{j=0}^{n_y-1}\left|r_{h_{e,i,j}}\right|}{\frac{3}{2}\sum_{a=0}^{n_s-1}\sum_{i=0}^{n_x-1}\sum_{j=0}^{n_{xy}-1}\sqrt{g_{i,j}}\,n_{a,i,j}\left(z_aT_e+T_i\right)}$$

$$\frac{\sum_{i=0}^{n_x-1}\sum_{j=0}^{n_y-1}\left|r_{h_{i,i,j}}\right|}{\frac{3}{2}\sum_{a=0}^{n_s-1}\sum_{i=0}^{n_x-1}\sum_{j=0}^{n_{xy}-1}\sqrt{g_{i,j}}\,n_{a,i,j}\left(z_aT_e+T_i\right)}$$

$$\frac{\sum_{i=0}^{n_x-1}\sum_{j=0}^{n_y-1}\left|r_{po_{i,j}}\right|}{\sum_{a=0}^{n_s-1}\sum_{i=0}^{n_x-1}\sum_{j=0}^{n_{xy}-1}\sqrt{g_{i,j}}\,z_a e n_{a,i,j}}$$